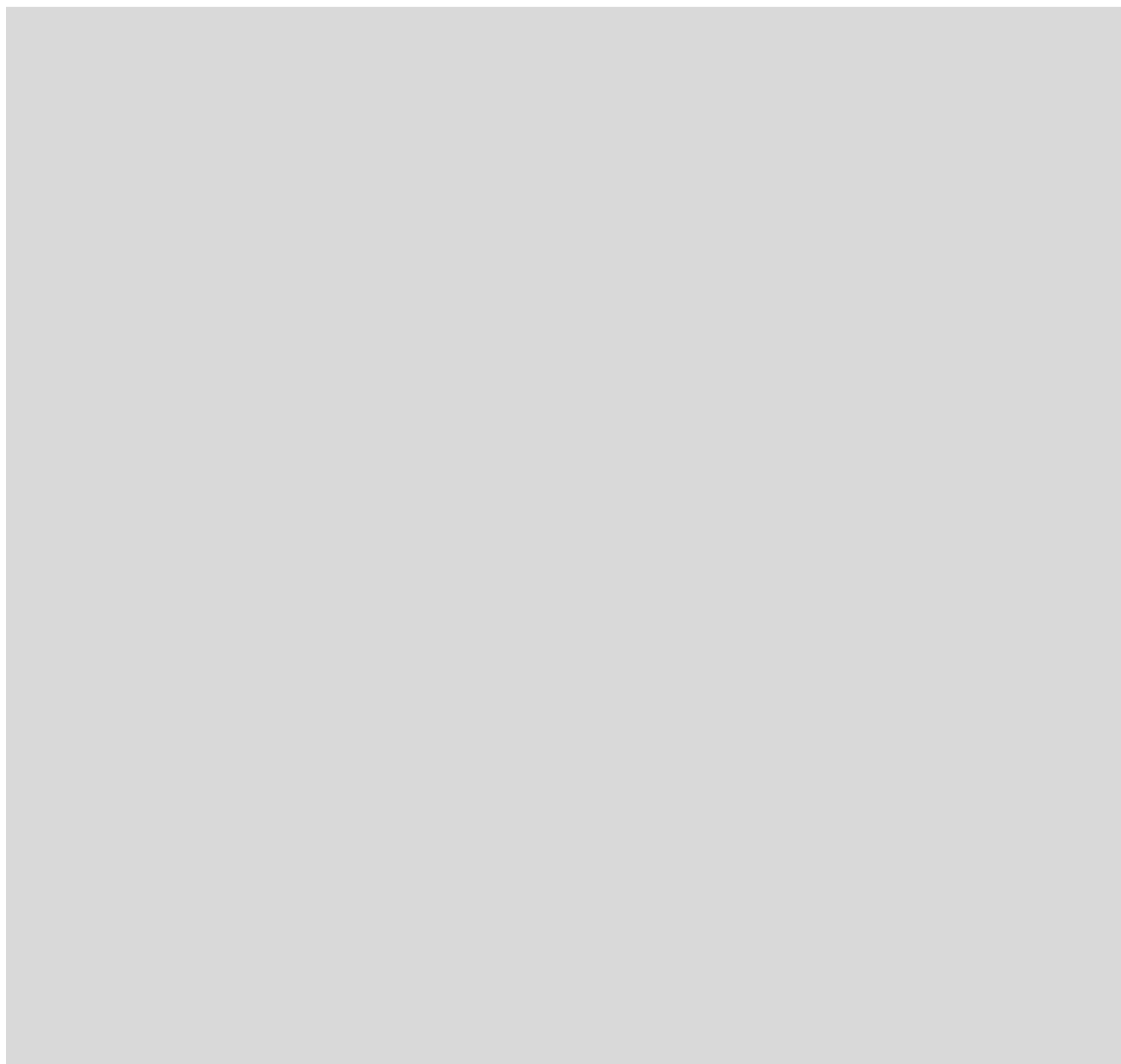


Typ1 osa / CC 220

Error messages



Version

104



BOSCH
Automationstechnik

Typ1 osa / CC 220

Error messages

1070 073 318-104 (95.03) GB

Z25 NJ / D25 ZA / T25 XA



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○ Foreword

General

The information on how to avoid and rectify errors is of a fairly general nature. It cannot account individually for all of the special conditions under which the control system will be used.

We therefore suggest that you also consult the user documentation of your control system during trouble-shooting. If in doubt, contact your authorized Bosch service point. If you need further information or have any serious problems with the control system that are not adequately described here or in the special user documentation, please contact Bosch service or our customer advisors. The addresses are provided inside the back cover of the manual.

Reference to the software version

These error messages refer to the control systems of the CC220 /320 and CC 200 / 300 series (software versions up to and including Z25NJ, D25ZA and T25XA).

NOTE!

The software version used in your control system is indicated in DIAGNOSTIC group operating mode when you press the DIAGNOSTIC CONTROL and SOFTWARE VERSION softkeys.

Reader target group

This manual addresses technically qualified personnel who have been specially trained and are familiar with CNC technology.

Trouble-shooting done by personnel who are not trained, or not adequately trained, may result in machine and equipment damage, loss of software, or even personal injury. It is therefore important that only properly trained personnel work with the control system and its individual components.

Qualified personnel as defined by the safety notes are people who have received instruction concerning CNC equipment, project planning and programming.

Please take note of our extensive range of training courses on these subjects. Training proposals are provided on the inside cover of the manual. Our training centre will be glad to respond to any queries you may have. The address is on the back of the manual.



○ Notes on safety

Please pay attention to the following safety issues:

CAUTION

Never plug in or disconnect cables/slot-in assemblies unless the system is switched off.

CMOS components can be destroyed by static discharge, so make sure you are earthed before you touch such components.

Do not open batteries using force, and do not try to recharge them or throw them onto fires. Dispose of spent batteries properly. Use only original replacement batteries.



Unless described otherwise, maintenance and repair work must never be carried out unless the system is switched off. If it is necessary to carry out measuring or testing work on the system when it is switched on, it is imperative to observe all safety and accident prevention regulations. Always use suitable electrical tools.



Repair/maintenance work on components of the CNC must be conducted only by Bosch service or by repair/maintenance agents authorized by Bosch.

All work on the electrical connection must be carried out by properly qualified personnel only, using suitable tools.



Unauthorized opening or removal of CNC components, or incorrectly executed repairs, can cause death, serious injury or considerable damage.



Removing/opening housings or a guard, or opening the switch cabinet, may expose equipment that carries dangerous voltages.



Emergency stop facilities must remain in effect in all modes of operation. Unlocking an emergency stop mechanism must never result in uncontrolled or undefined re-starting of the machine.

Explanation of the sticker symbols on the control system

The following stickers may be attached to the **actual control components** in order to draw your attention to certain things:



Warning, dangerous voltage!



Warning, battery hazard!



Components can be damaged by electrostatic discharge!



Disconnect power plug before opening!



Screws for connecting PE (protective earth) only!



Screen connection only!

1. Error messages, general

The control system displays error messages on screen as part of its safety functions. These messages are generated by such functions as hardware and software monitoring, input monitoring or program execution monitoring.

Errors always belong to one of the predefined error types:

- ☐ **System errors**
Error messages with the highest priority. A system error differs from other errors types in the way it is displayed (see 6. System errors).
- ☐ **P2 errors**
- ☐ **NC errors**
- ☐ **DNC errors**
- ☐ **Runtime errors**
- ☐ **Warnings**
- ☐ **Panel errors**
- ☐ **Customer area errors**

The error message always indicates the error type, error number and error text on the second and third row of the screen.

Example

P	1	NC0	P2 ERROR	116	MACHINE
.....		STOPPING	SERVO ERROR		13.8 18.45

At the same time, the most recent error messages are written out on the DIAGNOSTIC screen under ERRORS.

P	1	NC0	P2 ERROR	116	MACHINE
.....		STOPPING	SERVO ERROR		13.8 18.45
ERROR			PREPARATORY	STATUS	
SERVO ERROR					
1ST AXIS					
NC ERROR:					
PLC TIMEOUT					

It is not always possible to infer the exact cause of an error from the error message, since the error message is triggered by only one of a large number of possible causes.

The description attempts to specify the possible cause(s) for each error message, along with help on how to rectify the error.

There are various ways of clearing the error message from the screen, depending on which error occurred.

In the simplest case this is done by

- ✓ pressing a key
- ✓ returning to another level
- ✓ changing the mode of operation

These actions remove the error message from the header lines of the screen, but the error text is saved on the main DIAGNOSTIC screen.

DIAGNOSTIC mode provides three ways of clearing the error message from the header and in the main DIAGNOSTIC screen using softkeys.

Diagnostics

INTERVENTION FUNCTION				
DIAGNOSTIC CONTROL	DIAGNOSTIC MACHINE	INTERVENTION FUNCTION	SERVICE FUNCTION	
		HOME POSITION	CLEAR ERROR	CLEAR LOGIC

You can clear the error message using one of these functions, depending on the type and cause of the error. Sometimes the error may be rectified by restoring the control system to a controlled state.

Clear error

INTERVENTION FUNCTION				
		HOME POSITION	CLEAR ERROR	CLEAR LOGIC

Completely deletes the text of the error message. The current state is not changed.

Home position

INTERVENTION FUNCTION				
		HOME POSITION	CLEAR ERROR	CLEAR LOGIC

Completely deletes the text of the error message.

The control system displays the main DIAGNOSTIC screen. If a program is running or an input being made, for example, it will be aborted. Axis home positions are preserved, as is DNC.

Clear logic

INTERVENTION FUNCTION				
		HOME POSITION	CLEAR ERROR	CLEAR LOGIC

Clears logic; roughly equivalent to a hardware reset.

Completely deletes the text of the error message.

The control system powers up. Any running program is aborted and axis home positions are lost.

How to use this manual

The information on how to avoid and rectify errors is of a fairly general nature.

It cannot account individually for all of the special conditions under which the control system will be used. If in doubt, contact your authorized Bosch service point. The address of your local Bosch service point is provided inside the back cover of the manual.

In the following sections you will find the error messages sorted

- ✓ by **error numbers** – on pages 1–4 to 1–5
- ✓ **alphabetically** – on pages 1–16 to 1–28

in each case indicating the page on which you will find information on how to avoid and rectify the error.

Error messages that have been added since edition D3 are marked with a star (*).

The abbreviations used are explained on pages 1–29 and 1–30.

1.1 Error messages sorted by error numbers

This section contains all error messages, sorted by error numbers. The error messages of the panel (see page 1–15) are listed in alphabetical order, indicating the page number where you will find further information on how to avoid and rectify the error.

Error messages that have been added since edition D3 are marked with a star (*).

Error number/error message	Page
1 TEXT FILE MISSING	9 – 1
2 MACHINE PARAMETERS MISSING	9 – 1
3 CUSTOM. EPROM ON CODE AREA	9 – 1
1 CLEAR LOGIC	6 – 2
2 WATCHDOG COUNTER P2 NOT RUNNING	6 – 2
6 KS-DEF. PART DESTROYED	6 – 2
14 FILE TRANSFER ERROR	6 – 2
27 ACTIVE TABLE MISSING	6 – 3
50 TABLE NOT PRESENT	5 – 1
66 PROGRAM NOT PRESENT	7 – 1
67 INPUT ERROR	5 – 1/7 – 1
72 INCORRECT INPUT	7 – 1
73 INPUT INCORRECT	5 – 1/7 – 1
76 INVALID CHARACTER	5 – 1
77 ADDRESS EXISTS SEVERAL TIMES	5 – 1
78 FORMAT ERROR	5 – 2
78 INVALID FORMAT	5 – 2/7 – 1
78 FORMAT INVALID	5 – 3
79 UPGRADE LEVEL INACTIVE	4 – 1/5 – 4/7 – 2
80 SIGN ERROR	5 – 4
81 MAX. SUBROUTINE DEPTH	5 – 5
99 IS MAIN BLOCK SEARCH WANTED?	6 – 3
100 PROGRAM IN CONTROL	7 – 2
104 MEMORY FULL	4 – 1/5 – 5/7 – 2
105 ERROR IN FILE STRUCTURE	4 – 1/5 – 6/7 – 2
113 NO REF. POINTS	4 – 1/7 – 3
115 LIMIT SWITCH	2 – 1
116 SERVO ERROR	2 – 1
117 MARKER MISSING	2 – 1
118 MEASURING CIRCUIT ERROR	2 – 1
118 MEASURING SYSTEM FAULTY	2 – 1
125 BLOCK NON-EXISTENT	5 – 6/7 – 3
125 INADMISSIBLE COMPENSATION VALUE	6 – 3
134 TABLE IS ACTIVE	7 – 3
139 EMERGENCY STOP PENDING	2 – 1
146 PROGRAM IS ACTIVE	7 – 3
149 I/O DEVICE TIMEOUT	5 – 7/7 – 3
150 PARITY	5 – 7
165 ADDRESS INVALID *	5 – 7/7 – 3
172 LIMIT SWITCH	5 – 7

Error number/error message	Page
187 FILE HAS DFS IDENTIFIER	7 – 4
193 INVALID TANGENT	6 – 3
194 POSITION ERROR	2 – 1
195 NEGATIVE ROOT	6 – 3
196 INVALID REAL VALUE	6 – 3
196 ENCODER DEFECTIVE	2 – 2
197 F NOT PREALLOCATED	7 – 4
202 TIMING ERROR IN WAITFORRXR	6 – 3
203 NO INTERROGATION	6 – 3
206 FILE NOT DELETABLE	5 – 8
212 WRONG DIAG. VERSION	7 – 4
214 TIME TO BE SET	5 – 8
217 ERROR IN DIAGN. PRG.	7 – 4
223 CLEAR RESIDUAL DISTANCE G41/42	7 – 4
231 TIMING ERROR IN PLSINI	6 – 3
232 RAM ECHO ERROR IN PLSINI	6 – 4
234 NO RXR BIT FOR PANEL INTERRUPT	6 – 4
243 WRONG CODE NUMBER	7 – 4
253 WRONG INTERRUPT PCB	6 – 4
256 COUPLING BOARD RESET	6 – 4
260 NMI	6 – 4
261 ABORT	6 – 4
262 FPU TRAP	6 – 4
263 ILLEGAL OPERAND	6 – 4
264 SVC TRAP	6 – 4
265 DIV BY ZERO	6 – 4
266 FLAG TRAP	6 – 5
267 BREAKPOINT REACHED	6 – 5
268 TRACE TRAP	6 – 5
269 UNDEFINED OPCODE	6 – 5
270 ERROR: DUAL PORT TEST	6 – 5
271 ERROR: PARTY LINE TEST	6 – 5
272 ERROR: RAM TEST	6 – 5
273 ERROR: RELOAD TEXT	6 – 5
274 SVC NO. 6	6 – 5
275 HEAP OVERFLOW	6 – 5
276 SQRT OF NEG. ARGUMENT	6 – 6
277 BACKUP VERSION ACTIVE	6 – 6
280 ERROR: P2 INIT	6 – 6
281 TIMEOUT BY WAIT FOR P2	6 – 6
282 CRITICAL SECTION NOT ENDED	6 – 6
305 MEMORY FULL	4 – 1
323 EDIT ENABLE MISSING	4 – 1/7 – 4
324 PROGRAM ACTIVE	4 – 1
326 INADMISSIBLE FILE NUMBER	5 – 8
332 MAILBOX FULL	7 – 4
333 PANEL NOT SUITABLE	7 – 4
336 LINKING ABORTED	7 – 5
344 INPUT INCORRECT, AXIS RUNNING	5 – 8
346 TOO MANY AXIS MOVEMENTS	7 – 5
351 ALL LINK TAB. ACTIVE	5 – 8

Error number/error message	Page
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355 LINK ERROR	5 – 8/7 – 5
356 NO RUNTIME MEMORY	5 – 8
372 LSEC LOGIC	7 – 5
373 LSEC SYNTAX	7 – 5
374 LSEC GRID	7 – 6
375 LSEC OVERFLOW	7 – 6
377 NO CONTROL RESET POSSIBLE	7 – 6
379 FILE DESTROYED	5 – 8
386 LIMIT SWITCH 1ST AXIS	5 – 9/7 – 6
387 LIMIT SWITCH 2ND AXIS	5 – 9/7 – 6
388 LIMIT SWITCH 3RD AXIS	5 – 9/7 – 6
389 LIMIT SWITCH 4TH AXIS	5 – 9/7 – 6
390 LIMIT SWITCH 5TH AXIS	5 – 9/7 – 6
391 LIMIT SWITCH 6TH AXIS	5 – 9/7 – 6
392 LIMIT SWITCH 7TH AXIS	5 – 9/7 – 6
393 LIMIT SWITCH 8TH AXIS	5 – 9/7 – 6
394 INVALID COMPENSATION VALUE	5 – 9
395 INVALID ZERO SHIFT GROUP	5 – 9
396 INVALID FILE GROUP	4 – 1/5 – 9
397 FILE NAME FAULTY	5 – 9
398 PROTECTION FAULTY	5 – 9
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400 INVALID DISTURBANCE TABLE	7 – 7
401 NO DISTURBANCE TABLE	7 – 7
592 WRONG TOOL TYPE	7 – 7
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806 EPROM WRITE PROTECTED	7 – 7
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816 ERROR IN USER HEAP	6 – 8
817 LSV2 STATUS DATA FAULT *	6 – 8
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851 COPY FILE NOT PRESENT	6 – 8
950 STACK SUBRANGE EXHAUSTED	6 – 8
990 ILLEGAL MEMORY ACCESS	6 – 8
991 TASK MISSING IN NOTQ	6 – 8
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1.3 Abbreviations

ADR	ADDRESS
BABNET	BOSCH SYSTEM BUS NETWORK
BACKUP	LOADING BACK THE OPERATING SYSTEM
BPT	BREAKPOINT
CR / LF	END-OF-BLOCK IDENTIFIER
CLM	CLEAR MEMORY
COM	COMMUNICATION MEMORY
CTR	CONTROL RESET
DA	DIRECT EXECUTION (VIA DNC INTERFACE)
DFS	DEFINE STORAGE
DLG	DIALOG
DNC	DIRECT NUMERICAL CONTROL
EOF	END OF FILE
ETX	END-OF-FILE IDENTIFIER
EXP	EXECUTE EXTERNAL PROGRAM
ERR	ERROR
FSI	FILE STORAGE IDENTIFICATION
FPU	FLOATING POINT UNIT
GBA	GROUP OPERATING MODE
GRA	GRAPHICS
HEAP	MEMORY IN OPERATING SYSTEM
HP	MAIN PROGRAM
HS	HIGH SPEED
HW	HARDWARE
HW RESET	HARDWARE RESET (= CLEAR LOGIC)
IF	INTERFACE
ILL	ILLEGAL
IDU	JOB TYPE
KS	COMMUNICATION MEMORY
LF	LINE FEED
LSV2	DATA LINK PROTOCOL
LSV2—DNC	DNC WITH LSV2 PROTOCOL VIA V24/20 MA INTERFACE
LR	MASTER CONTROL COMPUTER
LSEC	LEADSCREW ERROR COMPENSATION
MDI	MANUAL DATA INPUT
MP	MACHINE PARAMETER
MPP	MACHINE PARAMETER PROGRAM

NPV	ZERO OFFSET
PLS	PANEL SOFTWARE
PTR	PAPER TAPE READER
P	PARAMETER
RTC	REAL-TIME CLOCK
RXR	RECEIVER READY
RXP	RECEIVE EXTERNAL PROGRAM
SK, SFK	SOFTKEY
SPIL	SOFTWARE PROGRAMMABLE INTERFACE LOGIC
SPG	SELECT PROGRAM
SVC	SUPERVISOR CALL
SW	SOFTWARE
TSU	TRANSPORT LAYER
UP	SUBROUTINE
WMH	MACHINE TOOL MANUFACTURER

2. P2 ERRORS

ERROR CAUSE		ERROR REMEDY
115 LIMIT SWITCH		
Machine position outside limit switch range.		* Move axis mechanically to limit switch range (permitted range); * Axis optimization (check overtravel and offset). For TR40 TC235 values X2,0; Y3,9: W8,0. * Approach position more slowly (axis overshoot); possibly reduce MPP P1007. * Check axis mechanisms. * Set SW limit switch tolerance higher; MPP 1507: > 0.050 mm.
116 SERVO ERROR		
Overtravel too great; setpoint limit exceeded. Tacho-generator interrupted. Setpoint interrupted and axis drifting.		* Control reset. * Check axis mechanisms. * Remedy cause.
117 MARKER MISSING		
Number of read-in increments of the encoder greater than defined in the MPP and still no zero pulse. Measuring system error; counted encoder pulses more than one revolution but no new marker. Encoder cable faulty (zero pulse line interrupted) Interface on the servo I/O board faulty.		* Check encoder. * Replace encoder cable. * Replace subassembly.
118 MEASURING CIRCUIT ERROR		
118 MEASURING SYSTEM FAULTY		
Encoder signal monitoring has responded. Measuring system removed or faulty.		* Check measuring system; clear logic.
Interface on the servo I/O board faulty.		* Replace board.
139 EMERGENCY-STOP PENDING		
Interface signal EMERGENCY-STOP pending.		* Remedy cause for emergency-stop; control reset
194 POSITION ERROR		
Number of read-in increments greater than or less than number of encoder lines; zero pulse present.		* Check encoder or cable; change cables round amongst each other.
Interface on the servo I/O board faulty.		* Check MPP P104.

ERROR CAUSE	ERROR REMEDY
196 ENCODER DEFECTIVE Overtravel of axis at standstill too great. Excessive axis drift. Measuring system does not supply any pulses. No axis movement, no measuring system pulses. Starting enable missing. Motor faulty.	* Perform offset adjustment. * Check measuring system. * Coupling between motor and measuring system loose? * Set drive enable. * Check motor.
2587 NO RESPONSE FROM AXIS PROCESS Axis processor card faulty.	* Check axis processor card and replace if necessary.
2588 WRONG CARD Servo card complement does not correspond to entry in MPP P9501.	* Correct entry in P9501 corresponding to servo card complement.
2602 NMI System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2603 ABORT System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2604 FPU-ERROR System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2605 ILLEGAL INSTRUCTION System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2606 SVC WITH ERROR System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2607 DIV. BY ZERO System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2608 FLAG TRAP System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2609 BREAKPOINT System error of the servo processor.	* Replace CP card. * Inform after-sales service.

ERROR CAUSE	ERROR REMEDY
2610 TRACE TRAP System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2611 UNDEF OPCODE System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2612 WRONG MODE IN CLOCK Defective P2 software or dual port faulty. Mode must be set to "2". This is permanently monitored by P2. If mode was switched, this acts as an indication that other data may be destroyed in the dual port RAM.	* Replace CP card. * Inform after-sales service.
2613 WRONG INTYPE IN CLOCK P1–P2 software versions do not match or data in dual port RAM destroyed.	* Replace CP card. * Inform after-sales service.
2614 RAM ERROR System error of the servo processor.	* Replace CP card. * Inform after-sales service.
2615 EPROM ERROR Defective P2 software. Error in EPROM test of processor P2.	* Replace P2 software on CP subassembly. * P2 software EPROMs correctly installed? * HW error on CP subassembly? * Inform after-sales service.
2616 WATCHDOG ERROR Computing operation of processor P1 too long (defective SW).	* Control reset (clear error message). * Note the last operator steps and inform the after-sales service.
2617 P2 INIT ERROR Defective or wrong P2-SW; Processor P2 has wrong data for start-up. HW error on CP subassembly.	* P2 software EPROMs correctly installed? * Replace CP.
2619 WRONG MODE IN LOOP Data in dual port RAM destroyed; dual port RAM overwritten.	* Reload software. * Change CP.
2620 VOLTAGE ERROR HS-I/O +24V missing for HS I/O board.	* Check +24V on the servo I/O board. * Check connector X31 on the servo I/O subassembly.

ERROR CAUSE	ERROR REMEDY
2624 INTERPOLATOR ABORT Travel command is present, but "Drive on" has been removed. Travel command is present, "Lock measuring system monitoring" is active via interface <u>and</u> the measuring system is defective.	* Control reset; check PLC program. * Replace measuring system.
2628 DISTANCE-ENCODED MEASURING SYSTEM <u>When using a distance-encoded measuring system:</u> A marker is not found - after 1 revolution with rotary transducer - or the 2nd cyclic (encoded) marker is not found. MPP 1514 is parameterized to "distance-encoded", but no distance-encoded measuring system is connected.	* Define MPP 1515, MPP 1517, MPP 104/105 accurately, possible reduce entry in MPP 502 (ref. point speed) * Connect distance-encoded measuring system.
2639 AXIS MONITORING After removing the travel command, the over-travel limit value entered in MPP 1012 is exceeded (after the waiting period specified in P 1013).	* Check entries in MPP 1012/1013. * Measuring system ok? * Inform after-sales service.

3. NC ERRORS

ERROR CAUSE	ERROR REMEDY
2661 UART OVERRUN ERR Data transfer speed not correct.	* Check baud rate. * Check handshake signals.
2662 OVERFLOW READ FIELD Loss of read-in data.	* Data input device faulty? Interface on the CP X11/20 mA faulty? Check interface on panel (V 24/20 mA). * Check baud rate.
2663 UART: ILL-COMMAND Inadmissible programming of the UART of the CNC interface (defective SW).	* Inform after-sales service.
2664 BUFFER BATTERY EMPTY The battery on the MEM board for buffering of the program memory (RAM) is exhausted.	* Replace battery and perform HW reset (refer to "Battery replacement" in "Interface conditions").
2664 CP/MEM BATT. VOLTAGE TOO LOW The battery on the CP/MEM board for buffering the program memory (RAM) is exhausted.	* Replace battery and perform HW reset (refer to "Battery replacement" in "Interface conditions").
2665 VOLTAGE FAILURE MACHINE I/O +24 V voltage missing on I/O subassembly.	* Check connector X10 (+24 V) on the I/O subassembly.
2665 VOLTAGE FAILURE 1ST I/O +24 V voltage missing on I/O assembly.	* Check connector X10 (+24 V) on the I/O assembly.
2666 UART FRAMING ERROR 20 mA current loop interrupted.	* Check connector on CNC, cable and connector on input device. * Data input device switched on? * Baud rate, number of stop bits correct?
2667 PLC-RAM ERROR BIT coupling board RAM is faulty.	* Replace NC-PLC coupling board.
2668 PLC PROGRAM < 2 ms PLC program requires less than 2 ms for one cycle. The PLC program must have a cycle length of min. 2 ms to permit proper access to the information on the bit coupling board.	* Increase PLC cycle time.

ERROR CAUSE	ERROR REMEDY
2669 PLC TIMING ERROR Cycle time longer than 10 ms. Faulty synchronization between PLC and NC No data exchange between CNC and PLC.	* Check the PLC program (endless loop). * Replace PIC board in the CNC.
2670 PLC TIME-OUT PLC not running (red LED "ZE stopped" lit) or running too fast. Calling alarm- or time-controlled modules can change the PLC cycle times. If the time defined in MP 6002 is exceeded, the error message is displayed. "Ready2" is briefly cleared.	* Check MPP P6002. * Check PLC program. * Connection PLC <—> NC correct? * Battery failure on ZE, NT, TZ? * Switch EDIT/MONITOR on ZE subassembly to "MONITOR" position.
2671 UART PARITY ERROR Parity error in data transfer X11/20 mA, CP, panel, terminal.	* UART module faulty; replace CP, replace panel control (panel/terminal). * Wrong data from paper tape reader; check paper tape (parity). * Input device faulty?
2672 RAM-PARITY ERROR RAM faulty. SW does not completely initialize RAM.	* Replace MEM board. * Clear logic.
2673 OVERFLOW KEYDAT Data loss due to excessively fast key operation; too many characters read by panel simultaneously.	* 1. Control reset * 2. Press keys more slowly. * ASCII keyboard faulty (key jammed).
2674 OVERFLOW POTINDAT Potentiometer buffer is full; loss of potentiometer data due to excessively slow processing.	* Panel control subassembly faulty? Replace subassembly. * Processor P1 too slow? Reload SW or replace CP subassembly. * Clear logic.
2675 OVERFLOW MTBINDAT MTB buffer is full. Panel sends too many data to CNC. Loss of MTB data due to excessively slow processing. Interface software wrong or faulty.	* MTB subassembly (046136—) faulty (TC 235 only); replace subassembly. * P1 processor working too slowly; reload SW or replace CP subassembly. * Interface processor or RAM faulty; replace CP subassembly. * Wrong interface software; replace EPROMs. * Clear logic.

ERROR CAUSE	ERROR REMEDY
2676 OVERFLOW CLOCKDAT Clock, Ident or Version buffer overflow.	* Panel control subassembly faulty? Replace sub-assembly. * Party-line SW faulty? * Clear logic.
2678 CHANNEL NOT OPEN Internal software error (interface cannot be opened). Log book output is aborted with mode change.	* Control reset; inform after-sales service if necessary. * Control reset.
2679 MEMORY DESTROYED One or more monitored memory areas have been destroyed.	* Load backup. * Inform after-sales service.
2680 UART ERROR 20 UART is defective.	* Inform after-sales service. — replace card —
2681 VOLTAGE FAILURE 2ND I/O 24V for the 2nd I/O card is not present.	* Check card or power supply connections

4. DNC ERRORS

ERROR CAUSE		ERROR REMEDY
79 UPGRADE LEVEL INACTIVE	If no messages are sent to the master control computer: KS tables not present.	* Apply KS tables; install word coupler.
104 MEMORY FULL	The LSV2 function "Reading-in" is aborted since insufficient memory space available.	* Provide more memory capacity and start reading-in again.
105 ERROR IN FILE STRUCTURE	An error has been discovered in the structure of the output file. The LSV2 function "Output" is aborted since an error has been found in the output file structure.	* Check file syntax. * Check file.
113 NO REF. POINTS	During activation of DA mode; reference points not approached.	* Travel to reference points.
305 MEMORY FULL	Memory space not sufficient for reading in files.	* Provide more memory capacity and start reading in again.
323 EDIT ENABLE MISSING	If no messages are sent to the master control computer: The edit enable is missing from the NC to permit a file to be deleted.	* Set edit enable.
324 PROGRAM ACTIVE	If no message is sent to the master control computer: A program is already active when a table is selected.	* Cancel program and repeat function.
396 INVALID FILE GROUP	If no messages are sent to the master control computer: The group is not defined within the NC.	* Change master control computer telegram.
399 FILE PROTECTED	If no messages are sent to the master control computer: Reading, writing, execution or deletion not permitted.	* Change file protection.

ERROR CAUSE	ERROR REMEDY
1131 FILE IN LIBRARY The file to be read in already exists in the library.	* Lock the file in the library or read in file with another number.
2010 FILE ALREADY EXISTS The file to be read in already exists in the control. The LSV2 function "Reading-in" is aborted since the file with the number of the DFS identifier already exists.	* Clear the existing file. * Read in file with another number.
2014 FILE TRANSFER ERROR The internal conversion routine reports an error (reading-in).	* Restart file transfer.
2220 FILE NON-EXISTENT If no messages are sent to the master control computer: Selected file does not exist in the NC. Error during output. Requested file does not exist.	* Transfer file to NC; change master control computer telegram. * Was search performed for correct file?
2366 V.24 ERROR USER A General error on interface X32*). — *) USER A = interface X32 USER B = interface X31	* HW reset.
2368 V.24 OVERFLOW USER A The UART on the V24 board reports overflow. More than the specified number of data bits were received.	* Set data format.
2370 V.24 PARITY USER A The UART on the V24 board reports a parity error after reception of a character.	* Set data format.
2372 V.24 PROTOC. ERROR USER A UART on the V24 board reports error after reception of a character.	* Set data format.
2374 V.24 INTERRUPTION USER A The UART on the V24 board reports an error after reception of a character.	* Set data format.
2376 RESTART ERROR V24 board reports error during start-up (re-initialization of the DNC interface).	* HW reset.

ERROR CAUSE	ERROR REMEDY
2377 EPROM ERROR V24 board reports error during testing of EPROM.	* Replace EPROM on the V24 board.
2378 V.24 RAM1 ERROR V24 board reports error of the interface RAM on the V24 board.	* Replace V24 board.
2379 V.24 RAM2 ERROR V24 board reports error of the interface RAM on the V24 board.	* Replace V24 board.
2380 V.24 COMMAND ERROR V24 board reports corrupt command from the NC-SW.	* Switch DNC off and then back on again.
2387 DNC TIMEOUT No further telegram was received within the monitoring period. A send telegram could not be sent to the master control computer during the monitoring period.	* File transfer was aborted: repeat file transfer. * Telegram to the master control computer was lost.
2389 CNC IN LOCAL MODE IF signal LOCAL/REMOTE not LOW.	* Switch NC to REMOTE.
2390 FCT NOT PERMITTED If no messages are sent to the master control computer: The group is not permitted in the NC for the selected function (e.g. activation of a diagnosis program).	* Change master control computer telegram.
2391 NO MEMORY ACCESS If no messages are sent to the LR: The memory is occupied at the NC by a different function.	* Abort function which is occupying the memory.
2392 FILE ACTIVE If no messages are sent to the master control computer: The file cannot be deleted because it is active.	* Cancel file, abort linking and editing.
2393 FCT CURRENTLY NOT POSSIBLE PLC has not yet acknowledged last remote job.	* Last job must first be acknowledged by the PLC; abortion of the last remote job.
2395 NC RUNNING A program is already being executed when DA mode is activated.	* Set FEED HOLD, cancel program.

ERROR CAUSE	ERROR REMEDY
2397 MONITOR/SIMULATION ACTIVE DNC cannot be switched off with active monitor/ simulation.	* Abort diagnosis program, then switch off DNC.
2398 INVALID FILE NO If no messages are sent to the master control computer: The No. is not permitted for the selected function within the group (e.g. reading in the file).	* Change master control computer telegram.
2399 READING-IN ACTIVE The CC200/300 refuses a request of the DNC computer for the LSV2 function "Automatic" at the initiative of the CC200/300 since READING-IN is active via a different interface.	* Abort reading-in and restart automatic execution.
3540 TOOL MAGAZINE When selecting a program via LSV2, not all programmed tools are in the magazine.	* Tool completely before selecting the program.
3541 MANUAL CHANGE NOT PERMITTED Program selection via LSV2: More tools are programmed than the magazine will hold.	* Modify the program so that the tools are enough.
4641 DNC HARDWARE ERROR Interface board reports an HW error during the DNC operation (LSV2-DNC). The BAB subassembly reports an HW fault.	* Check interface board. * HW reset; if this is not successful, replace the subassembly.
4642 BAB: ADDRESS ERROR The specified destination address cannot be accessed.	* Connect the corresponding user with the bus.
4644 BAB: NO CENTRAL CONTROL UNIT The BAB user does not have any connection to the BAB central control unit.	* Check functioning of central control unit. * Check bus for interruptions.
4650 BAB: CRC ERROR This error message can occur only for the BAB central control unit (control cannot be central control unit). It was not possible to transfer a telegram after 2 attempts.	
4651 TSU-BAB BLOCKED The BAB subassembly does not enable the interface RAM.	* Switch the DNC on and off.

ERROR CAUSE	ERROR REMEDY
4653 BAB STATUS PORT Status port copy was not updated. An interrupt was "swallowed".	* No remedy necessary, DNC continues running.
4654 UNDEF. A IDU (INTERNAL) Unknown LSV2 function within the CNC	* Use defined functions.
4658 BAB: USER NOT READY User cannot accept a telegram since his RAM is still blocked by a preceding telegram, for example.	* Create memory. (Error message can occur only with BAP central control unit).
4659 PLC JOB KS DATA LSV2: Invalid address in the job list of the PLC. LSV2: Specified group in the job list of the PLC does not exist. LSV2: Specified table in the job list of the PLC does not exist.	* Correct job list. * Correct job list. * Correct job list.
4660 KS SEND TAB BLOCKED LSV2: Access to KS table is blocked.	* Enable KS table.
4662 OVERFLOW TA TABLE No free space in the reference table.	* Switch DNC off and on.
4663 TSU: IDU FAULTY The transport layer (TSU) has received an undefined job type.	* The TSU rejects the one job; communication continues.
4664 TSU: REC. TELEGRAM TOO LONG A receive telegram exceeds the permitted length. The partner exceeds the permitted LSV2 telegram length during telegram transmission.	* No remedy necessary. The corresponding telegram is rejected by the TSU. * Compare the LSV2 telegram length of the control and partner and set to valid value.
4665 TSU: REC. SEQUENCE NO. The sequence number contained in the telegram header is invalid.	* No remedy necessary. The corresponding telegram is rejected.
4666 TSU: SEND SEQUENCE NO. The send sequence number is not plausible. Loss of a telegram.	* No remedy necessary. Communication continues.
4667 TSU: INVALID TSUOPC. The TSU opcode contained in the telegram header is unknown.	* The telegram is rejected. Communication continues.

ERROR CAUSE	ERROR REMEDY
4668 TSU: SENDING BLOCKED Telegrams cannot be sent because the receiver does not acknowledge.	* The error is caused by the receiver.
4669 TSU: SEND RESET Unknown reason in a reset telegram for the send channel.	* Communication software is not compatible. Computer and control cannot communicate with each other.
4670 TSU: RECEIVE RESET Unknown reason in a reset telegram for the receive channel.	* Computer and control cannot communicate with each other.
4671 TSU: RESTART Undefined reason in a restart telegram.	* Computer and control cannot communicate with each other.
4672 TSU: INVALID CHANNEL NUMBER A send job of the application layer contains an invalid channel number.	* The error is caused by the application layer.
4682 INDIVIDUAL DISABLE ACTIVE At least 1 axis is disabled. As a result of this the job cannot be executed.	* Cancel axis disable.
4691 FUNCTION BLOCKED BY PLC Job was refused by the PLC.	* Try again, change PLC prog.
4693 ERROR PENDING IN CNC NC-START is not possible if an error (except DNC error).	* Remedy error, abort function.
4695 FUNCTION ALREADY ACTIVE Function is already active.	* Wait.
4698 WRONG PROGRAM DA mode cannot be activated since the sent program does not correspond to the requested one.	* Send requested program.

ERROR CAUSE	ERROR REMEDY
4708 READING-IN: FILE STRUCTURE LSV2 automatic: DNC computer sends file reading-in end without correct end in the file data. LSV2 reading-in: The following errors are established while reading in the data: parity error, E-code error, syntax error in the file structure, invalid compensation value, invalid zero offset group, file name error in DFS identifier, file protection error in DFS identifier, invalid SPG or CLM identifier, various errors for W, S, K files.	* Check file. * Check file.
4709 READING-IN: PROTOCOL ERROR Incompatible function code used during reading in via LSV2 by master control computer. LSV2: The DNC computer does not keep to the prescribed protocol for the reading-in function.	* Check protocol on computer side. * Check protocol.
4710 OUTPUT: PROTOCOL ERROR Internal error during output. The master control computer sends a defective output request (protocol) to the control. LSV2: The DNC computer does not keep to the prescribed protocol for the output function. LSV2: The DNC computer sends a faulty request telegram (wrong file number, group or syntax) for the output function at the initiative of the control.	* Restart output. * Check the request telegram from the master control computer. * Check protocol. * Restart the function "File output".
4711 FILE-I/O UTILITY ERROR Internal LSV2 system error, for the functions READING-IN, OUTPUT and CLEAR. Internal LSV2 system error, for the function DIR output.	* DNC control reset for the LSV2 functions READING-IN, OUTPUT and CLEAR (otherwise switch off DNC and then switch back on again). * DNC control reset (otherwise switch off DNC and then switch back on again).
4713 OUTPUT ABORT: COMPUTER The DNC computer aborts the LSV2 output function by DNC control reset.	* Restart output.
4714 READING-IN ABORT: COMPUTER The DNC computer aborts the LSV2 reading-in function by a DNC control reset.	* Restart reading-in.

ERROR CAUSE	ERROR REMEDY
4715 READING-IN: WRONG FILE The number or group of the file transferred by the computer is not that which was requested. A file cannot be read in since the number contained in the FSI identifier already exists. The DNC computer sends an invalid file number in the DFS identifier in the LSV2 function "Reading-in an L FILE". The FSI identifier group stored in the file does not agree with the file group (reading-in). The DNC computer sends an incorrect group in the DFS identifier for the LSV2 function "Reading-in". The DNC computer sends a corrupt request telegram (wrong file number, group or syntax) for the LSV2 function "Reading-in" at the initiative of the CC200/300.	* Check protocol on computer side. * Clear the existing file. * Restart reading-in. * Check the group under which the file is known in the master control computer. * Restart reading-in. * Restart reading-in.
4716 DATA AFTER FILE END Characters were still received during reading-in after M30 or M2. The DNC computer sends data after the file end for the LSV2 functions "Reading-in" or "Automatic".	* The file is created without the characters after M30 or M2.
4717 NO ACCESS TO DNC RAM The CC200/300 cannot enter an LSV2 telegram on the V24 board since the board does not enable the internal RAM.	* Check V24 board.
4718 TELEGRAM CODE UNKNOWN An unknown code is discovered on the user level when decoding a telegram. The DNC computer sends an unknown LSV2 telegram identification.	* No remedy necessary; transfer continues. * Check telegram identification.
4719 LINK FAULTY An LSV2 telegram cannot be sent by the CC200/300 since the link to the computer is broken.	* Check connecting cable.

ERROR CAUSE	ERROR REMEDY
4720 COMPUTER NOT READY TO RECEIVE An LSV2 telegram cannot be sent by the CC200/300 because the computer does not accept the telegram (neg. acknowledgement).	* Check DNC computer.
4721 DNC UNDEF ERROR CODE V24 The V24 board signals an internal error to the CC200/300 (not further defined).	* Switch DNC off and then back on again; check V24 board.
4722 TELEGRAM TOO LONG DNC computer sends an excessively long LSV2 telegram (telegram length > than defined in the corresponding machine parameter).	* DNC computer must correct telegram length.
4723 CLEAR FILE ERROR Various errors in the LSV2 function "Clear file". – File group not defined – Syntax error – File or file group delete-protected – File does not exist – File is active – File is being used – File is in library	* Remedy cause of fault; repeat function.
4724 PLC NOT INITIALIZED LSV2 initialization with the PC600 has not functioned with PC600 coupling.	* Check PLC block.
4725 PLC DATA TRANSFER The PLC acknowledges an LSV2 telegram from the DNC computer negatively.	
4726 PLC SEND TELEGRAM PLC wishes to send an excessively long telegram to the DNC computer (telegram length > than defined in corresponding MP). PLC uses an NC telegram identification.	* Check PLC block.
4733 PLC BLOCKING RECEPTION PLC no longer accepts any LSV2 telegrams from the DNC computer.	* Check PLC block.
4737 DNC CONTROL RESET BY COMPUTER File transfer aborted by master control computer.	* Restart file transfer.

ERROR CAUSE	ERROR REMEDY
4739 PROTOCOL ERROR The DNC computer sends an unauthorized telegram for the LSV2 function "Automatic".	
4740 ACKNOWLEDGE DNC MESSAGES Buffer for new messages is full since the last message is still acknowledged.	* Do not acknowledge received messages.
4750 STATUS TRANSFER IMPOSSIBLE No memory space can be reserved for buffering the status messages when the DNC is switched on.	* Increase USER-HEAP (SKONF).
4751 NO STATUS MESSAGE There is no space for buffering the status data when a status message is generated.	* No remedy necessary.
4752 NO OPERATION MESSAGE When a state which must be signaled is changed, no operation message can be generated since the previous message is still pending for transfer.	* No remedy necessary.
4753 MZA RECORDING The PLC does not react to the MZA request of the NC.	* Change the PLC program.
4754 DATA BLOCK W/O DATA The DNC has sent an empty data block during the function "Direct execution" or "Reading-in".	* Check the output logic of the DNC computer.
4760 BLOCK TRANSFER ACTIVE The CC200/300 refuses a request of the DNC computer for the LSV2 function "Automatic" at the initiative of the CC200/300 since block transfer is already active.	* Abort block transfer and restart automatic execution.
4761 SYNTAX ERROR The DNC computer sends a corrupt request telegram (wrong syntax) for the LSV2 function "Automatic" at the initiative of the CC200/300.	* Restart automatic execution.
4763 AUTO. ABORT: COMPUTER The DNC computer aborts the LSV2 function "Automatic" by a DNC control reset.	* Restart automatic execution.

ERROR CAUSE	ERROR REMEDY
4804 KS ACCESS BLOCKED Reading-in of KS tables is currently blocked. Output of KS tables is currently blocked. LSV2 function "Reading-in/output" is aborted since access to the W file is blocked (PLC access). LSV2 function "Reading-in" is aborted since the DNC computer has sent corrupt KS data.	* Start reading-in again later. * Start output again later. * Restart reading-in. * Enable PLC access. * Tool file corrupt.
4810 KS-DEF: TEXT LENGTH The LSV2 function "Reading-in" is aborted since the DNC computer has sent an error in the definition part of the KS data.	* KS table definition part corrupt.
4811 KS-DEF: NUMBER OF GROUPS The LSV2 function "Reading-in" is aborted since the DNC computer has sent an error in the definition part of the KS data.	* KS table definition part corrupt.
4812 KS-DEF: ADDRESSES The LSV2 function "Reading-in" is aborted because the DNC computer has sent an error in the definition part of the KS data.	* KS table definition part corrupt.
4813 KS-DEF: STATUS TEXT The LSV2 function "Reading-in" is aborted because the DNC computer has sent an error in the definition part of the KS data.	* KS table definition part corrupt.
4814 KS-DEF: GROUP TEXT The LSV2 function "Reading-in" is aborted because the DNC computer has sent an error in the definition part of the KS data.	* KS table definition part corrupt.
4815 KS-DEF: SUB-TABLE The sub-table to be read in is undefined. The LSV2 function "Reading-in" is aborted because the DNC computer has sent an error in the definition part of the KS data.	* Check sub-table number. * KS table definition part corrupt.
4816 KS-DEF: SYNTAX The LSV2 function "Reading-in" is aborted because the DNC computer has sent an error in the definition part of the KS data.	* Tool file corrupt.
4817 OUTPUT ACTIVE "Output" is active when DA mode is activated.	* Abort output.

Note:

If a transfer error occurs, all data which are received up until the point when the error is detected are still processed. The control then reinitializes the interface.

When an interface hardware error is detected, the current DNC function is aborted and the DNC interface is deactivated.

5. RUNTIME ERRORS

ERROR CAUSE	ERROR REMEDY
50 TABLE NOT PRESENT It was attempted to activate a table which does not exist in the NC block with G22 K... or V... One of the G-codes G154 – G159 or G254 – G259 was programmed in the NC block. The corresponding group is not activated in the machine parameter P4501. Compensation or zero shift table does not exist. The KS table specified in the CPL TD command does not exist.	* Change part program or create corresponding table. * Change part program or change P4501 "Number of zero shift groups". * Create or reload table. * Change CPL program or create or reload desired table.
67 INPUT ERROR No dwell time (F) was programmed in the block for the dwell time in spindle revolutions (G104). The F-value is missing in the NC block with G4 (dwell time).	* Always specify a dwell time (F) in the block with dwell time in spindle revolutions (G104). * Add F-value in the NC block.
73 INPUT INCORRECT A block was selected for a block search in which thread cutting (G33) is modally active.	* Only select blocks in which thread cutting is not active for the block search.
76 INVALID CHARACTER Invalid character in the part program. Block skip symbol "/" is not programmed in the first column. A letter is programmed for which the type is set to 1 (error) in the MP 9000.	* Read in program again. * Change part program. * Change part program or MP.
76 INVALID CHARACTER The character "\" (backslash) is detected at the start of a block in the part program during active program testing, although no break-point was set for this block.	* Remove the "\" symbol.
77 ADDRESS EXISTS SEVERAL TIMES An NC address which must be programmed only once in the NC block (settable via MP 9000) is programmed several times. G875, G876, G877, G878 is in one block with another G code. Further G codes in addition to the couple command are programmed in one NC block.	* Change the part program or MP 9000. * Change the NC program. * Change the NC program

ERROR CAUSE	ERROR REMEDY
78 FORMAT ERROR Read-in format for F, M or S word is invalid.	* Program in valid format.
78 INVALID FORMAT Variable notation (VAR=33, G[VAR]) is not permitted for thread cutting (G33).	* Program G33 without CPL-variable.
Variable notation (VAR=75, G[VAR]) is not permitted for measuring probe input (G75).	* Program G75 without CPL-variable.
An asynchronous axis is programmed whose address letter is assigned an incorrect type value in the machine parameter program (P9000).	* Change MPP. * Address type may have the following values: 2, 3, 5, 6, 7, 8, 10, 11
An auxiliary function address has a word content 2147483647 or –2147483648.	* Change part program.
An auxiliary function address has the type 12 (MP 9000).	* Change MP.
The NC word content of an auxiliary function address is greater than allowed by the number of integer positions (machine parameter P2007).	* Change part program or MPP.
An NC address of the type 12 (machine parameter P9000) "String" has a CPL expression as a variable address which does return a string.	* Change part program.
The permitted number of digits of an NC word (machine parameter P9000) has been exceeded.	* Change part program.
A blank NC word has been programmed, i.e. no digits. e.g.: N10XY100.	* Change part program.
An NC address of the type 2, 5, 8, 11 Integer or 12 Str (machine parameter P9000) has been programmed with a ".", i.e. as a real value.	* Change part program.
A G-word with variable address is interpreted as G23, G24, G70, G71 or G92, e.g.: G[92].	* Change part program.
A G98 is programmed in the NC block and the modally present preparatory function is not G0 or G1.	* Change part program.
The K or V addresses possess the type 3, 6 or 10 (MPP P9000) in the NC block with G22 and the word content is 2147483647 or –2147483648. (Possible only with variable address and CPL expression).	* Change part program.

ERROR CAUSE	ERROR REMEDY
The K or V address is type 4 (MP P9000) in the NC block with G22.	* Change MP.
Incorrect transfer parameter- of the type "Real" – for subroutine call.	* Check and correct transfer parameters of the corresponding subroutine call.
Incorrect transfer parameter – of the type "Integer" – for subroutine call.	
Spindle speed not declared as real or integer.	* Change MP 9019.
Retraction angle lies between 60 and 90 degrees.	* Change retraction angle.
Retraction angle < 0 degrees.	* Change retraction angle.
An auxiliary function subroutine call is programmed with variable address, for example A[10] subroutine call defined with A10 in the machine parameter P4007.	* Change auxiliary function code.
An M subroutine call is programmed with variable address (machine parameter P4006). A G subroutine call is programmed with variable address (machine parameter P4002).	* Change M code.
Wrong format for compensation value table.	* Change G code.
Programmed Hirth axis value is invalid.	* Check and change compensation values.
A value is programmed via a switching function which does not correspond to the format defined in the MPP.	* Change part program. * Enter prog. value again in correct format.
The format of the address M is set to 1, 4, 9 or 12 in the MPP P9000.	* Change MP.
Rotary axis for tangential tool guidance is programmed with active circular interpolation.	* Change part program.
A P or Q subroutine call is programmed with variable address or more than one subroutine call is present in the NC block.	* Change NC block.
78 FORMAT INVALID	
Variable notation was used when programming G876, G877.	* Change NC program.
Programming of CPL variables (VAR=52, G/ VAR/) is not allowed when programming G51, G52.	* Program G51, G52 without CPL variable.

ERROR CAUSE	ERROR REMEDY
79 UPGRADE LEVEL INACTIVE Max. 3 synchronous axes may be programmed in interpolation mode (G1, G73, ..) in the COCOM version. Call of a non-activated upgrade level. Invalid G-code programmed. The CPL code word is reserved. The INP#(0,...) or PRN#(0,...) command was used although the upgrade level CPL WITH PROGR. GRAPHICS is not active. The INKEY command was used although the upgrade level CPL WITH PROGR. GRAPHICS is not active. CPL graphics commands were used although the upgrade level CPL WITH PROGR. GRAPHICS is not active. The reserved call TXT was programmed. A reserved function call was used. Error during reading-in: S files are not permitted since no SPIL or PIC 200 is present. Inadmissible jump out of application; software error. SPG identifier programmed. AXIS COUPLING upgrade level was programmed. Application bit not set. G875 in CNC with servo 8-card prog. No limit switch range programmable with G876, G877 was defined or no servo-i available. Switch over spindle with G201/G202 with 2nd spindle not applied.	* Program max. 3 synchronous axes in interpolation mode (G1, G73...). * Inform after-sales service; retrofit upgrade configuration level. * Change part program. * Other variable name. * Change CPL program. * Inform after-sales service, retrofit upgrade configuration level. * Change variable name. * Change variable name. * Apply SPIL or PIC 200. * Inform after-sales service. * SPG identifier is not permitted. * Adapt application. * G875 only possible with servo-i. * Define range in MP6200 or use servo-i. * Apply upgrade level "2nd spindle".
80 SIGN ERROR Positive for an NC address with type 5 or 6 (machine parameter P9000). Negative value is programmed for integer or positive real. Negative feed programmed.	* Change part program. * Enter positive feed.
81 MAX. SUBROUTINE DEPTH Nesting of the subroutines is too deep.	* Change subroutine call structure.

ERROR CAUSE	ERROR REMEDY
104 MEMORY FULL Warning for copying: File is to be copied, but no memory space is available. The simulation memory via which the data relevant to simulation are forwarded to the output task is created upon entry into NC block simulation if this memory does not yet exist. Memory space insufficient. There is no longer sufficient memory space for the OPENW command in order to create the desired file. There is no longer sufficient memory space for automatic expansion of a file opened for writing in the PRN#(...) command. Error during reading-in: There is not sufficient memory space available for reading in a file. The memory space is not sufficient for creation of a KS table. There is no space available to copy the NC program for "program optimization". There is no space to create a DA buffer. Not enough memory to record image. Not enough memory for diagnostic file when starting D28.	* Create memory space by deleting files and/or control reset; perform function again. * Reduce the simulation memory or delete other files. * Create memory space by deleting files and/or control reset. * Create memory space by deleting files and/or control reset. * Create memory space by deleting files and/or control reset. * Possibly increase MP 6005. * Create memory space by deleting files and/or control reset. * Create memory space by deleting files and/or control reset. * Create memory space by deleting files and/or control reset. * Create memory space by deleting files and/or control reset.

ERROR CAUSE	ERROR REMEDY
105 ERROR IN FILE STRUCTURE Error during reading-in: The number of the DFS identifier is larger than 14 digits. Error in the structure of a D file. Error in the end identifier (M02, M30, ETX). Two successive DFS identifiers are separated without an end identifier. SPG identifier programmed. Compensation values occur during reading in of a geometric compensation table which lie outside the permitted value range. An invalid checksum occurs for a PIC 200 program. Error during reading-in of D files in Intel hex format: CR and LF missing at the start of a line. Trying to convert invalid characters from ASCII to HEX. More than 16 data bytes occur in one line. The first position of the status byte is not 0. Checksum in a line is not correct. First four bits of base segment record contain values not equal to 0. The value for the status byte is not equal to 0, 1, 2. The object code format code is corrupt for PIC200. Image file contains false characters, lines that are too long, or has a corrupt structure.	* Correct number. * Check D-file. * Program M02 <LF> or M30 <LF> or <ETX> at the end of a file. * Program M02 <LF> or M30 <LF> or <ETX> after the first file. * SPG identifier is not permitted. * Change invalid compensation values of the read-in data. * Check and correct checksum. * Insert CR and LF. * Convert only values from 0..9 and A..F. * Maximum of 16 data bytes permitted in one line. * First position of the status byte must be 0. * Correct checksum. * First four bits of the base segment record must be 0. * The status byte may contain only 0, 1, 2. * Check object code format. * Regenerate image file or re-read via data I/O.
125 BLOCK NON-EXISTENT Entered block number does not exist in this file.	* Enter existing block number.

ERROR CAUSE	ERROR REMEDY
149 I/O DEVICE: TIME-OUT Error during reading-in/output: Data not arriving. Connection faulty. Reading-in via DCR: All data has already been transferred. The error message is issued at the tape end.	* Check device connection, device setting and interfaces.
150 PARITY Error during reading-in Parity error occurs if – EIA occurs with even parity sum. – ISO occurs with odd parity sum.	* Data device supplies incorrect times; check data link. * Set parity correctly.
165 ADDRESS INVALID An axis movement was programmed in the same block as the dwell time in spindle revolutions (G104). – NO AXIS or – AN ASYNCHRONOUS AXIS or – A ROTARY AXIS or – AN AXIS VALUE $\neq 1, -1$ was programmed for switch-over of drilling cycle and length compensation (G78). A letter is programmed for which the type is set to 9 (MPP P9000). An axis address is programmed in the NC block with G4 dwell time. An address is programmed which is not N, P, K or V in the NC block with G22. An axis address is programmed in the NC block with G22. Axis address not in selected level. S address programmed in G33 block. Address of coupled slave axis was programmed while coupled.	* Do not program travel in the block with dwell time in spindle revolutions (G104). * Switch-over of drilling cycle and length compensation must occur only to a synchronous linear axis in positive or negative direction. (Example: G78 Y–1). * Change part program or MPP. * Change part program. * Change part program. * Change part program. * Switch level. * Remove S-address from G33 block. * Change NC program
172 LIMIT SWITCH Traverse path programmed which is outside the programmed limit switch. Programmed value for the rotary axis for tangential tool guidance is outside the limit switch range.	* Program traverse path within the limit switch range. * Program axis value within the limit switch range.

ERROR CAUSE	ERROR REMEDY
206 FILE NOT DELETABLE Error during reading-in: CLM identifier if file group is not P, V, K or G.	* Only delete files from the permitted file groups P, V, K or G.
214 SETTING CLOCK Date or time cannot be read with the CPL Date/Time command.	* Problems with party-line? * The time is currently being entered.
326 INADMISSIBLE FILE NUMBER Error during reading-in: File number not permitted. 1. Number < 1 or > 999999999. 2. Reading-in blocked for this file (e.g. L1).	* Correct file number.
344 INPUT INCORRECT, AXIS RUNNING Trying to switch over to the C-axis, but one axis still possesses a travel command.	* Initiate switch-over from the PLC only when no travel command is pending.
351 ALL LINK TAB. ACTIVE It is not possible to create a new link table because all ten non-permanent link tables are active.	* Create additional permanent link tables and restart linking.
354 LINKING ALREADY ACTIVE A program has been selected for which no link table has yet been created; a link process for another program is already active.	* Abort linking and restart program.
355 LINK ERROR A program has been selected for which a link process has already been performed; an error has been established.	* Establish the type of the error in NC mode under SK "Table selection"; remedy error and reselect program.
356 NO RUNTIME MEMORY There is no longer sufficient memory in order to allow the program to run.	* Create memory space by deleting files and/or control reset; call program again.
379 FILE DESTROYED The block end identifier "Line feed" and the file end identifier ETX are missing in the last block of a file when a string is read with the I NP#(...) command.	* Check opened file.

ERROR CAUSE		ERROR REMEDY
386	LIMIT SWITCH 1st AXIS	
387	LIMIT SWITCH 2nd AXIS	
388	LIMIT SWITCH 3rd AXIS	
389	LIMIT SWITCH 4th AXIS	
390	LIMIT SWITCH 5th AXIS	
391	LIMIT SWITCH 6th AXIS	
392	LIMIT SWITCH 7th AXIS	
393	LIMIT SWITCH 8th AXIS	
	Limit point outside the limit switch ranges.	* Change part program or limit switch.
394	INVALID COMPENSATION VALUE	
	Error during reading-in of zero shift files: Invalid zero shift value.	* Change zero shift value.
395	INVALID ZERO SHIFT GROUP	
	Error during reading-in of zero shift files: Invalid zero shift group value.	* Change zero shift group.
396	INVALID FILE GROUP	
	Error during reading-in: Invalid file group in the DFS identifier.	* Correct file group.
397	FILE NAME FAULTY	
	Error during reading-in: File name in the DFS identifier is longer than 30 characters.	* Shorten file name.
398	PROTECTION FAULTY	
	Error during reading-in: File protection in the DFS identifier contains more than 4 characters or is not R, W, E and D.	* Correct file protection; may contain a maximum of 4 characters: R, W, E and D.
399	FILE PROTECTED	
	Trying to activate a table which is protected (type of protection E) in the NC block with G22 K... or V....	* Change part program or protection E.
	Trying with OPENR to open a read-protected file.	* Change CPL program or file protection.
	Trying with OPENW to open a write-protected file.	* Change CPL program or file protection.
	A TTY interface to be written by the PRN#(...) command is already open.	* Terminate parallel access, e.g. by control reset.
	The main program or a subroutine is pro- tected against execution.	* Change protection or check call structure.

ERROR CAUSE		ERROR REMEDY
696 MEMORY FULL	Graphic memory is full. Part program memory is full.	* Store graphics and call again with "Old part" (Trumpf). * Delete part programs which are not currently required – control reset.
786 DEAD ZONE BLOCKED	Incorrect position in the claw dead zone without "claw testing" having taken place.	* Change part program.
787 INVALID C-VALUE	C-axis is programmed with active circular interpolation.	* Change NC program.
788 INADMISSIBLE A-VALUE	Programmed acceleration outside the limit values defined in the MPP.	* Change programming or limit values.
789 G6 W/O AXIS INFO	Acceleration programming without input of acceleration values.	* Specify acceleration values for at least 1 axis.
789 TAP TOO SMALL	Tap too small for desired thread.	* Select suitable tap.
790 NO E-WORD PROGRAMMED	M22 programmed without E-word.	* Program E-word.
791 INCORRECT CIRCULAR PATH PROGRAMMING	M22/25 end point not on circular arc. Circle without I, J.	* Change program.
792 INVALID E-WORD	Traverse path per stroke greater than permitted traverse path, e.g. X10 E15 M22	* Program E > 0.01 or < 999.99.
793 INVALID H-WORD	H > 999 or < 1	* Program H < 999 or > 1.
794 H WITHOUT TRAVERSING INFO	H-word programmed without traverse path.	* Enter traverse path; e.g. X500 H10 M22.
795 LIMIT POSITION WRONGLY CALCULATED	Limit position is not correct.	* Inform after-sales service.
796 INVALID LIMIT POSITION	Limit position below claws.	* Programming error in part program in the case of externally created programs.

ERROR CAUSE	ERROR REMEDY
797 MAXIM. 4 CLAWS More than 4 claws were defined during claw test.	* Correct program.
800 ERROR IN PLC PROGRAM A syntax error has been found in the PLC program. Error during reading-in: In S-file (only PIC200 program). An error has occurred in INTEL-HEX format.	* Correct the PLC program. * Check and correct format.
1080 LINK TABLE NOT PRESENT A CPL program was started in machine, data I/O, diagnosis or compensation mode for which no link table exists.	* Create link table for the corresponding CPL program.
1131 FILE IN LIBRARY Trying to open a file with OPENR or OPENW which already exists in the library. It is attempted to read in a file which exists in the library and is not locked. The file specified in FIXB already exists in the P group of the library as a cleared program.	* Change CPL program or block file in the library. * See above. * Change file no. in FIXB or lock program in library.
1137 LOAD KS TABLE Reading-in or output is active. Error during reading-in of KS tables. Reading-in cannot be started during copying of KS tables.	* After reading-in or output, copy KS tables with diagnosis program D10. * Read in again via diagnosis program. If the error occurs again, the cause of the error is described more precisely by a new error message. * Control reset; restart KS table copying operation.
1217 SWITCH-OVER NOT PERMITTED Test operation without total block or auxiliary block. Rapid retraction of the C-axis prog. and trying to switch back to the spindle without canceling rapid retraction of the C-axis. G95, 33, 330, 104 is active and trying to switch to the C-axis. Trying to switch over to the C-axis in the block search without traveling to the reference point. Switch-over to C-axis without reference point and C-axis requires reference point. Trying to switch over to the C-axis via a switching function with act. program.	* Activate total block and auxiliary block. * Switch off rapid retraction of the C-axis with G99 and program new retraction positions without C axis. * Switch off invalid G-functions before switch-over to C-axis. * In a block search, program all switch-over operations with travel to reference point (MP 7605). * Set all switch-over operations with MP 7605 or MP 1510 for C-axis to YES. * Switch-over via switching function is not possible with active program.

ERROR CAUSE	ERROR REMEDY
1218 INVALID PROGRAMMING A travel movement of a synchronous axis is contained in the switch-over block C-axis spindle/spindle-C-axis. The C-axis is programmed in turning operation. A spindle M-function or an S-word is contained in the switch-over block.	* Write switch-over M-code alone in a block. * Delete C-axis programming in turning mode. * Write switch-over M-code alone in a block.
1228 OSCILLATING AXIS ACTIVE G993 is active and the oscillating axis is programmed by means of its address letter.	* Change part program or activate G994.
1236 G92 IS ACTIVE G52 (set workpiece zero) must not be used with active G92 (set actual value).	* Cancel G92.
1242 KS ACCESS BLOCKED PLC access takes longer than 10 ms.	* Change PLC program (enable access for NC).
1258 DRIVE ON ERROR Drive on was cancelled without permission of NC.	* Change PLC timing (refer to "Interface conditions").
1259 STARTING NOT POSSIBLE Safety cut-out during input of subsequent switching functions. Clear residual travel with active safety cutout.	* Control reset, program cancellation or renewed program selection. * Control reset, program cancellation or renewed program selection.
1365 INSUFFICIENT CHECK POINTS Trying to execute a SPLINE block without a sufficient number of check points having been programmed. Switch-over from SPLINE interpolation to a different interpolation mode has taken place. Only two check points were available for block preparation for spline interpolation up to then; it was not possible to execute the SPLINE block.	* At least three check points must be available for block preparation in order to permit execution of a spline arc. * At least three check points must be available for block preparation in order to permit execution of a spline arc.
1366 INVALID G CODE A G-function was programmed which must be used only with active spline interpolation. An invalid G-function was programmed with active spline interpolation.	* Activate spline interpolation or remove G function previously. * Remove corresponding G-function from the program.

ERROR CAUSE	ERROR REMEDY
1367 INVALID CHECK POINT Two successive check points are identical with active spline interpolation. Three successive check points are in inadmissible configuration on a straight line.	* Successive check points must differ from each other. * The third check point must lie on the connecting line from the first and second check point in positive direction at most.
1368 INVALID WORKING AREA Trying to use the laser head in a prohibited area.	* Control reset/cancellation and correction of the faulty program.
1418 ALTERNATIVE TOOL Error in management of alternative tools; 1. Alternative tool and alternative compensation group do not match or there is no suitable alternative tool for the programmed tool. 2. An endless chain may also be programmed in the combination of alternative X tools. An invalid configuration may possibly have been generated via CPL (TC/TD command). 3. The software may also be defective.	* Check the alternative tool combination in the tool editor. * Check the alternative tool combination in the tool editor. If combination OK: control reset and program the T again in the switching function.
1427 INVALID X ZERO SHIFT Zero shift not approached and G491 is programmed.	* Approach zero shift before G491 is programmed.
1428 +X SIDE WRONG (MPP 7513) Machining in the positive X area is not possible during face machining (refer to MPP P7513).	* Change entry in MPP P7513 or change program.
1429 –X SIDE WRONG (MPP 7513) Machining in the negative X area is not possible during face machining (refer to MPP P7513).	* Change entry in MPP P7513 or change program.
1440 G92 OR ZERO SHIFT GROUP CHANGED A zero shift is programmed while G491 is active.	* Change NC program.
1441 C-AXIS NOT ACTIVE C-axis machining is not programmed.	* Program C-axis machining (G20C...Z...).
1449 ZERO SHIFT OR COMP. CHANGE FOR G491 G491 is active. Offsets are changed.	* Offsets must not be changed with active G491. Change NC program.

ERROR CAUSE	ERROR REMEDY
1451 T WITH ACTIVE G491 A tool change was programmed with active face machining (G491).	* Change NC program; program T-word before G491.
1456 TABLE CHANGE WITH ACTIVE G491 Comp. and zero offset table switchover with active G491.	* Change NC program; no comp. or zero offset tables must be switched when G491 is active.
1457 FALSE PARAMETER IDENTIFIER False CPL parameter was transferred. Reading a tool compensation table; invalid character read.	* Change the CPL program. * Correct the table.
1458 PERMITTED VALUE RANGE EXCEEDED Error during reading-in: The permitted data range is checked when a compensation table is read in. It is checked whether the respective limit values are exceeded, e.g.; limit values of machine parameters for T, LX, LZ, DX, etc.	* Correct the compensation table or the MP (P7501..P7510) before reading in again.
1459 INVALID STRING/TRANSFER ERROR Error during reading-in: A character was found during reading-in of a compensation table which was not expected at this position. e.g.: "KO" was recognized, whereby this must be followed by numeric information.	* Repeat reading-in (transfer error?); correct table.
1463 TOOL NO. NOT DEFINED Error during reading-in: Before the parameters of a tool are written in a line, the No. of the tool must be specified. (e.g.: WZ 03...)	* Repeat reading-in (transfer error?); correct table.
1464 COMPENSATION NO. NOT DEFINED Error during reading-in: Before the parameters of a compensation group can be written in a line, the No. of the compensation group must be specified. (e.g.: KO 12 LX ...).	* Repeat reading-in (transfer error?); correct table.
1466 NO LZ VALUE It is established in block preparation that a tool (T-word) has no length offset in Z. LZ = 0 means tool carrier point!	* Measure tool? Reenter tool? Change programming?

ERROR CAUSE	ERROR REMEDY
1472 NO COMPENSATION GROUP ACTIVE CPL: TC command A TC command without specification of the compensation group was programmed but no compensation group is active (T0)!	* Activate tool (compensation group) or change TC command.
1473 NO TOOL ACTIVE CPL: TD command A TD command without specification of the tool was programmed but no tool is active (T0)!	* Activate tool or change TD command.
1474 G92 X.. NOT PERMITTED Programming error: G94 is not permitted in X	* Change programming.
1475 TOOL CHANGE ONLY WITH G40 Programming error: G41, G42 The cutter radius compensation must be switched off before the tool change.	* Program G40 or correct the program.
1480 ALT. COMPENSATION GROUP INVALID Programming error: G41, G42 The cutter radius compensation must be switched off before the tool change.	* Program G40 or correct the program.
1498 TOOL DATAW1-W4 MISSING KS tables W1-W4 not written.	* Complete tool tables in the KS table.
1499 INVALID PROGR. IDENTIFIER Program identifier MACHTYP does not correspond to the permitted values.	* Use permissible values.
1500 INVALID TOOL CALL SD function called with invalid T (K, D) numbers. Number is outside the permitted value range. Selected feed rate too high.	* Value range: T 1–99 K1–999999999 D 100–99912 * Check feed in MPP.
1507 INVALID CHANGE POSITION No tool change is permitted at the calculated change position.	* Check position of the Y-axis and parameter P3 in M06.
1535 INVALID TURNING TOOL Machining with the turning tool is not possible owing to the correct cutting edge/ milling cutter position.	* Error in the tool table or select different turning tool.

ERROR CAUSE	ERROR REMEDY
1536 ZERO OFFSET NOT APPROACHED An offset is switched on, but not approached with programmed G491.	* Change NC program; at least one traversing block must be programmed for offset approach before G491.
1542 BLOCK SEARCH NOT ALLOWED Block search to a block with active G491 is not allowed.	* Block search only possible to blocks with G490.
1543 G108 ACTIVE G491 activated while G108 is active.	* G491 only allowed with G9.
1544 DISTANCE FROM ZERO POINT Trying to pass the coordinate origin closer than is allowed.	* Change NC program. Determine "allowed" distance using SD (104).
1550 G33: LEAD NOT ALLOWED No lead value programmed in the G33 block. Lead of the cutting axis is negative. Max. permitted lead exceeded. Min. permitted lead not reached. No retraction possible since retraction angle same as lead angle.	* Program at least one lead value. * Enter positive lead. * Reduce lead. * Increase lead. * Change retraction point or retraction angle.
1551 SPINDLE SPEED NOT PERMITTED Programmed speed outside the permitted range.	* Correct spindle speed.
1553 G33: RETRACTION NOT PERMITTED Retraction movement with sign change in main axis path part. Retraction movement with sign change in auxiliary axis path part.	* Change retraction point or retraction angle. * Change retraction point or retraction angle.
1553 RETRACTION NOT POSSIBLE Retraction movement with sign change in main-axis path. Retraction movement with sign change in auxiliary axis path.	* Change retraction point or retraction angle. * Change retraction point or retraction angle.
1554 ALL VARIANTS No F value is programmed or the value is < 0.0001 in G98 block "Rapid retraction". The programmed feedrate for G95 is too high.	* Change NC program with respect to F-value. * Change programmed feedrate.

ERROR CAUSE	ERROR REMEDY
1554 FEED NOT PERMITTED No F-value is programmed or the F-value is < 0.0001 in the G98 block "Rapid retraction". Feed becomes negative owing to excessively large neg. feed change. Permitted feed exceeded (main axis). Permitted feed exceeded (auxiliary axis). Permitted feed not reached. Programmed feed for G95 too large.	* Change part program. * Correct feed change F. * Reduce spindle speed. * Reduce spindle speed. * Increase spindle speed. * Change programmed feed.
1555 NO G96 POSSIBLE Thread cutting in conjunction with G96 (constant cutting speed).	* Deactivate G96.
1555 NO DIST. PROGRAMMED Thread-cutting in conjunction with programmed G96.	* Deactivate G96.
1556 SYNC. NOT POSSIBLE Spindle synchronisation is not possible: – One or both spindles are under automatic control (M5/M19) – Spindles rotating in different directions – One spindle is in test mode – One or both spindles have gear stage acknowledgement – The 100% speed of the lead spindle cannot be reached by the passive spindle in the selected gear stage – Block search in a phase with synchronous operation	
1557 SPINDLE PROG. NOT PERMITTED Invalid spindle progr. when switching on or with synchronisation active.	* G192/292, G193/293, movement M codes, gear stage M codes and S values are not allowed.
1558 SPINDLE INTERF. ERROR Invalid interface modification with synchronous mode active.	* Only "0 Volts lateral", "pot internal/external" and "pot 100%" can be changed.
1559 M30 WITH SYNC. ACTIVE Main program end reached with spindle synchronisation active.	* Control reset. Change part program; i.e. cancel synchronisation before M30.

ERROR CAUSE	ERROR REMEDY
1577 BARRIERS EXCEEDED A barrier limiting line is intersected by the cutting edge tip in a traversing movement which was processed by block preparation. Barrier transgression is recognized at the following elements. — Barrier transgression at the chuck. — Barrier transgression at the steadyrest. — Barrier transgression at the tailstock.	1. Check traversing movement. * Is the end point of the traversing movement inside a barrier? * Does the programmed curve cut a barrier? * Do not execute traversing movement; change program. 2. Barrier points incorrectly measured? An excessively large safety distance places restrictions on machining. * Remeasure barrier points. 3. Check tool measurement. * Is the correct tool programmed? * Is the compensation group compensation value/group correct?
1985 ADDRESS “S” MISSING G875 or G890 programmed without “S”.	* Change part program. In G875, “S” is the distance of the height-controlled axis (P8006 /53/) until the measurement is finished.
1986 ADDRESS “S” OR “F” MISSING G890 programmed without “S” and “F”.	* Change part program.
2005 CLOSE BRACKETS MISSING The closing bracket is missing in the NC block after an opening bracket.	* Change part program.
2008 FILE NON-EXISTENT Error during reading-in: The file sought by way of the DFS identifier does not exist.	* Correct file sought?
2010 FILE ALREADY EXISTS The program already exists when DA mode is activated. Trying to read in a file during data transfer which already exists in the part program memory. The file specified in FIX already exists in the part program memory.	* Set FEED HOLD; cancel program. * Change the file no. in FIXB.
2015 INVALID LEVEL The selected level contains at least one axis which is not applied in the system. G20 with more or less than two axis addresses. Level changed with coordinate transf. activated.	* Change selected level. * Program G20 with two axis addresses. * Change NC program.

ERROR CAUSE	ERROR REMEDY
2020 RADIUS > 5000 mm The workpiece radius is too large for cylinder jacket machining.	* Reprogram G20 with radius.
2028 INVALID WEIGHTING FACTOR Weighting factors are programmed in the G114 block which are outside the limit values defined in the MPP (P1014, P1015).	* Program different weighting factors or change MPP.
2030 CLG VALUE INVALID Programmed closed-loop gain (CLG) outside the limit values defined in the MPP.	* Change programmed values or limit values.
2031 G14 WITHOUT AXIS INFO Closed-loop gain programming without input of CLG values.	* Program CLG value for min. one axis.
2040 G2/3: F/R INVALID Feed too large referred to radius.	* Reduce feed.
2041 INVALID CIRCLE PARAMETER No radius (R) must be used with active quadrant programming. The interpolation parameters have a negative sign. 1. Circle parameters do not match (e.g. distance start point-end point > 2) 2. Radius > 100 m. 3. Full circle not permitted for radius programming. 4. No center-point coordinates programmed. Can also occur in limit cases if the radius input is inexact.	* Program the circle center point I, J; I, K or J, K. * Program the interpolation parameters I, J, K as relative values without sign. * Correct circular-path programming. * Check MPP 9903, 9912.
2042 G2/3: INVALID NUMBER OF AXES Too many axes programmed.	* Correct part program.
2043 G5: DIRECTION UNDEFINED No tangential approach direction can be calculated.	* Correct circular-path programming.
2044 G5: RADIUS TOO LARGE Calculated radius too large for tangential approach.	* Correct part program.
2045 G5: AFTER HELICAL G5 programmed after helical interpolation.	* Correct part program.

ERROR CAUSE	ERROR REMEDY
2046 G5: AFTER LEVEL CHANGE G5 programmed after level change.	* Correct part program.
2048 UNEQUAL FACTORS Different scaling factors exist and circular traversing is taking place.	* Change scaling factor or do not traverse circular blocks (G2, G3).
2049 LEAD TOO LARGE Lead greater than 82 degrees. No circular component in the movement.	* Reduce lead. * Check NC block. * Traverse linear axis with G 1.
2095 ERROR IN ECODE Error during reading-in: Incorrect E-code value. E-code inputs not permitted for S (PIC), G, K, W, V and D files.	* Correct E-code values. * Do not use an E-code.
2098 PERM.-VARIABLE BLOCKED Access by CPL to permanent variable unsuccessful since permanent variable is blocked by the PLC.	* PLC must reset interlock flag for permanent variables.
2101 PROHIB. WORKING AREA AXIS 1 2102 PROHIB. WORKING AREA AXIS 2 2103 PROHIB. WORKING AREA AXIS 3 2104 PROHIB. WORKING AREA AXIS 4 2105 PROHIB. WORKING AREA AXIS 5 2106 PROHIB. WORKING AREA AXIS 6 2107 PROHIB. WORKING AREA AXIS 7 2108 PROHIB. WORKING AREA AXIS 8 For G25, G26: The indicated axis is not in the permitted working area or leaves the permitted working area with linear interpolation.	* Check program.
2109 PROHIB. WORKING AREA CIRCLE For G25, G26: The permitted working area is left with circular interpolation.	* Check program.
2110 WORKING AREA MAX. 3 AXES For G125, G126: More than 3 axes are programmed in the definition of the prohibited working area.	* Check program.
2116 NO AXIS COUPLING POSSIBLE Trying to activate axis coupling although the two axes are in different positions.	* Change NC program.

ERROR CAUSE	ERROR REMEDY
2120 TOO MANY BLOCKS SUPPRESSED With active cutter radius compensation, too many blocks without traversing movement in the active level. More than 6 blocks without traversing movement are programmed in total before or after a traversing block. The CPL command WAIT was programmed.	* Change part program.
2121 LEVEL CHANGE ACT. COMP. A different level is activated via level switch-over (G17..G20) with active cutter radius compensation. With face end interpolation active, tool radius compensation is deselected without movement.	* Change part program. * Change part program. Move out tool radius compensation before G490 (level change) with at least one programmed traverse movement.
2122 COLLISION !! With active cutter radius compensation: Tool offset radius is greater than full circle radius. The software is not in a position to calculate a corrected contour, but cannot recognize any of the other errors. With active collision monitoring: The original contour would be violated by execution of the corrected contour.	* Change part program or select a different compensation value.

ERROR CAUSE	ERROR REMEDY
2123 NO INTERSECTION It is not possible to calculate an intersection point between two contour elements offset by the cutter radius compensation value with active cutter radius compensation. The error may occur if: Two elements, line – circle or circle – circle enclose an acute angle and the corrected elements do not have a common intersection point. With active collision monitoring, if three elements (combination of straight lines and/or circles) form a pocket, whereby the 2nd element can no longer be executed owing to the compensation value and there is no intersection point between the 1st and 3rd elements. Owing to computing inaccuracies in calculation of the corrected elements if the elements possess tangential transitions.	* Change part program or select a different compensation value.
2125 INVALID COMPENSATION VALUE An invalid G-function was programmed with active spatial tool offset G43/44. Incorrect compensation value. Radius compensation value too large or too small. An invalid value is assigned to a variable in a KS table with a CPL TD command. The maximum permitted value is defined by the format agreed for the variable in the definition part of the KS table.	* Change G-functions in the part program. * Check and change compensation values. * Change radius compensation value in part program or change corresponding MP. * Change CPL program, possibly incorporate check.
2126 SEVERAL ZERO SHIFTS ACTIVE Wrong parameters in CPL command FXC.	* Check and change parameters in the CPL command in the part program.
2131 COMP. ENTRY WITH CIRCLE With active cutter radius compensation: The 1st traversing movement after activation of the cutter radius compensation logic via G41/G42 is a circular movement.	* Change part program. Program linear movement.

ERROR CAUSE	ERROR REMEDY
2132 COMP. CANCEL WITH CIRCLE With active cutter radius compensation: The 1st traversing movement after deactivation of the cutter radius compensation logic via G40 is a circular movement.	* Change part program. Program linear movement.
2133 CHANGE G41/42 WITH CIRCLE With active cutter radius compensation: The 1st traversing movement after a change from G41 to G42 or from G42 to G41 is a circular movement.	* Change part program. Linear movement after change.
2134 COMP. ACT. AT PROGRAM END With active cutter radius compensation: The cutter radius compensation is still active at the end of the main program. The main program cannot be executed up to the end.	* Change part program; G40 at main program end.
2136 G-ADDR INVALID FOR G41/42 With active cutter radius compensation: One of the following G-addresses is programmed: – Set zero point G92. – Inch-metric switch-over G70 to G71 or G71 to G70. – Switch-over of the active level with G17 ... G20. – Thread cutting G33. – Travel to reference point G74, G174, G274. – Rapid retraction G98/G99.	* Change part program.
2136 G-ADR INVALID FOR G41/G42 G32 is programmed with active cutter radius compensation.	* First cancel cutter radius compensation.
2137 SINGLE BLOCK WITH G41/42 Program testing in “single block” is active with active cutter radius compensation.	* Select subsequent block or single step.
2138 NO COMPENSATION ACT. CPL command TC programmed without previously activating a compensation table. Wrong parameters in CPL command FXC.	* Check and change part program. * Check and change parameters in the CPL command in the part program.

ERROR CAUSE	ERROR REMEDY
2140 MODAL SUBROUTINE IS ACTIVE It is possible to jump to a subroutine or jump back into the main program in direct execution mode only if no modal subroutine is active: A subroutine is called although a modal subroutine is active. A subroutine is quit although a modal subroutine is still active.	* Deactivate modal subroutine with G80.
2142 NO M2 OR M30 An M2 or an M30 is missing in the main program or in a subroutine.	* Insert M2 or M30.
2144 MEMORY IS OCCUPIED The memory has not been released for the link table which is to be created. The memory is occupied by other I/O actions (reading in etc.).	* Enable memory and restart link. * Abort the other I/O action.
2145 TOO MANY VARIABLES The memory requirement for the global variables or for the local variables of a subroutine is greater than 32767 bytes.	* Change CPL program; DIM instruction possibly too large.
2150 INSUFFICIENT MEMORY Each PRN instruction within a DLG sequence requires memory space in the system heap. However, the memory is full. There is not sufficient memory space in order to create a link table.	* Enable system heap again with CLEAR LOGIC. * Create memory space by control reset or by deleting files and restart linking.
2151 VARIABLE TYPE WRONG A variable name is present several times with different types.	* Check variable name in CPL program.
2152 SUBROUTINE NON-EXISTENT A subroutine call is programmed for which there is no subroutine. Trying to read the local data of a subroutine which is not present in the link table. Fix was called with a non-existent file no. Go block function/re-entry was selected. A subroutine is specified in MPP P4014. There is either a subroutine call or tool change call in the main program, but the subroutine does <u>not</u> exist.	* Change part program or create subroutine. * Delete link table, perform linking again. If this error occurs during runtime, an internal software error exists. * Change Fix no. or read in missing image file via data I/O. * Create subroutine with the part program number which is defined in the MPP or set MPP P101 to "0".

ERROR CAUSE	ERROR REMEDY
2153 NESTING TOO DEEP An expression in a CPL instruction is nested too deep.	* Reduce expression by means of local variables.
2154 VARIABLE NOT OCCUPIED Read access is performed to a variable which has not yet had a value assigned.	* Check CPL program.
Write access is made to a field variable for which no DIM instruction has yet been run through.	* Check CPL program.
The index limits of a real field variable were not observed.	* Change field variable.
A variable is still undefined, i.e. no value was assigned to it previously.	* Change CPL program.
2155 WRONG INDEX The index limits of a field variable were not observed.	* Change field variable.
2157 SELF-CALL PROHIBITED The main program or a subroutine calls itself or is called again in a deeper nesting.	* Change subroutine call structure.
2158 INVALID STATEMENT It was attempted to write the external zero shift values with CPL command FXC or the PLC is not connected via a word coupler.	* Check part program and change if necessary or check connection of PLC.
One of the commands OPENW, OPENR, CLOSE, REWRITE, INP# and PRN# (file access) is programmed in a DLG sequence.	* Change CPL program.
The EOF command is used in a DLG sequence.	* Change CPL program.
The “=” symbol is missing after one of the following commands: IC, MIC, FXC, TC or TD.	* Change CPL program.
A CPL instruction is not yet terminated at the end of the program.	* Change CPL program.
The last block in a program is a CPL block and it does not end with a <LF>.	* Change CPL program.
2159 THEN MISSING The “THEN” is missing after the “IF” in front of the Boolean expression.	* Change IF instruction.

ERROR CAUSE	ERROR REMEDY
2160 ENDIF MISSING The “ENDIF” is missing after the instruction part of “IF...THEN” or “ELSE”. Particularly also if “ENDIF”s are missing at the end of the program.	* Check IF instructions and correct if necessary.
2161 WRONG BRANCH DESTINATION The GOTO instruction is followed by something other than an NC block number, an integer constant, a real constant or a CPL label.	* Change GOTO instruction.
2162 UNTIL MISSING The “UNTIL” is missing after the instruction part of “REPEAT”. Particularly also if “UNTIL”s are missing at the program end.	* Check REPEAT instruction.
2163 TO MISSING “TO” is missing after “FOR” after the instruction.	* Check FOR instruction.
2164 NEXT MISSING The “NEXT” is missing after the expression following the “TO”. Particularly also if “NEXT”s are missing at the program end.	* Check FOR instruction.
2165 DO MISSING The “DO” is missing after “WHILE” after the Boolean expression.	* Check WHILE instruction.
2166 END MISSING The “END” is missing after “WHILE...DO” after the instruction part. Particularly also if “END”s are missing at the program end.	* Check WHILE instruction.
2167 = MISSING The “=” is missing for write access of the COM-command. The “=” symbol is missing after a variable, a feed variable, a variable after “FOR” or after “COM” (write access).	* Change CPL program. * Change corresponding instructions.

ERROR CAUSE	ERROR REMEDY
2168 INADMISSIBLE VARIABLE	
The parameter "Program name" is not of the type "String" for the OPENW command.	* Change OPENW command.
The expression on the right side is not of the type integer, real or double-precision real for write access to the COM command.	* Change COM command.
The subroutine parameter is of the type "Integer", but the transfer parameter is of the type "Real".	* Change type of transfer parameter or type of subroutine parameter.
Incorrect variable type or inadmissible compensation value in CPL command TC.	* Check and change part program.
Variable in CPL command possesses wrong type.	* Check and change part program.
An expression of the type "Real" is assigned to a CPL TD command.	* Change type of expression on the right side to "Integer".
A global variable does not start with a letter or number after the "#".	* Change name of global variable.
A field variable is programmed without corresponding DIM instruction.	* Insert DIM instruction.
No variable follows "FOR".	* Change FOR instruction.
No field variable follows "DIM".	* Change DIM instruction.
The expression on the right side of the IC instruction is not of the type "Boolean".	* Change IC instruction.
An expression is assigned to the MIC command which is not of the type "Integer".	* Change MIC command.
A second DIM instruction was programmed with the same variable, but with different memory requirements.	* Change corresponding DIM instruction.

ERROR CAUSE	ERROR REMEDY
2169) MISSING	
The bracket is not closed [“”] after the program identifier “TTY1 “ or “TTY2” for the OPENR or the OPENW command.	* Change OPEN command.
The bracket is not closed [“”] after the program identifier “TTY” and the file No. for the OPENR or the OPENW command.	* Change OPEN command.
The bracket is not closed [“”] after the program identifier or after the file No. for the OPENR command.	* Change OPENR command.
The bracket is not closed for one of the following CPL instructions: CIR, COL, FIL, GMD, GWD, INP, LIN, MWD. The bracket is not closed for a field variable within a PRN instruction.	* Change CPL program.
The bracket is not closed [“”] after the file No. for the OPENR command.	* Change OPENR command.
The bracket is not closed [“”] after the program identifier for the OPENR command.	* Change OPENR command.
There is no length or the bracket is not closed for the OPENW command after the program identifier or the file No. or there is no program name or the bracket is not closed [“”] after the length or the bracket is not closed after the program name [“”].	* Change OPENW command.
The bracket is not closed [“”] or there is no comma (“,”) after the first parameter of the CLOSE or REWRITE command.	* Change corresponding command.
Too many parameters are specified or the bracket is not closed [“”] for the COM command.	* Change COM command.
The bracket is not closed: – in the parameter list of a CPL instruction. – in a field variable. – in a nested expression.	* Change CPL program. * Change field variable. * Change expression.
A “]” follows the CALL subroutine call.	* Change CPL program.

ERROR CAUSE	ERROR REMEDY
2170 INVALID EXPRESSION	
A CPL expression is programmed in an NC word of the address types 2, 3, 5, 6, 7, 8, 10 or 11 (MP 9000) with variable address which returns a value other than integer, real or double-precision real, e.g.: X[A\$].	* Change CPL expression in part program.
A CPL real expression is programmed in an NC word of the address types 2, 5, 7, 8 or 11 (MP 9000) with variable address which is 2147483647 or 2147483648.	* Change part program.
The specified value for asynchronous axes is outside the internal representation range.	* Specify a permitted value.
No number follows the “.” of a real constant.	* Supplement real constant.
There is no variable or no bracket is opened [“(”] after “NOT”.	* Change NOT expression.
CPL commands are used in an expression which do not return any value, e.g.: A%+OPENR...	* Change expression.
No bracket is opened [“(”] after a “NOT” in the first parameter of WAIT.	* Change WAIT instruction.
An expression of the type “Boolean” or “String” is programmed in a variable address of an NC block.	* Change NC block.
Something else is programmed as well as the expression in a variable address of an NC block, e.g.: X[A,].	* Change NC block.
Something else is programmed in a block as well as the CALL instruction.	* Change CPL block.
A character field was used for programming a variable address.	* Change CPL expression.
There is no integer constant after the CALL subroutine call.	* A variable subroutine call is not permitted. Insert integer constant!
2171 – NOT PERMITTED	
A Boolean or string expression has a negative sign.	* Change CPL expression.

ERROR CAUSE	ERROR REMEDY
2172 TYPE MUST BE BIT The expression after “NOT” is not of the type “Boolean”. The comparison with the IC command is not of the type “Boolean”. The condition in the IF, WHILE or REPEAT instruction is not of the type “Boolean”.	* Change NOT expression. * Change WAIT instruction. * Change corresponding instruction.
2173 INVALID OPERAND An expression was used for the “+”, “-”, operation which is not of the type “Integer”, “Real” or “Double-precision real”. An expression is used for the “AND”, “OR” or “XOR” operation which is not of the type “Boolean”.	* Change expression. * Change expression.
2174 INVALID RELATION An expression is used for the “<”, “>”, “<=”, or “>=” operation which is not of the type “Integer”, “Real” or “Double-precision real”. For the “=” or “<>” operation, the first operand is of the type “Integer”, “Real” or “Double-precision real” and the second of the type “Boolean” or “String”, or the first V operand is of the type “Boolean” and the second is of a type which is not “Boolean” or not “NUL”, or the first parameter is of the type “String” and the second of a type which is not “String” or not “NUL”.	* Change expression. * Change expression.
2175 INVALID ASSIGNMENT The variable on the left side of the type “Double-precision real” and the data type (3rd parameter) is not “D” for read access of the COM command. The variable on the left side is of the type “Real” and the data type (3rd parameter) is not “R” for read access of the COM command. The variable on the left side is of the type “Integer” and the data type (3rd parameter) is not “I”, “W” or “B” for read access of the COM command. The variable on the left side is not of the type “Integer”, “Real” or “Double-precision real” for read access of the COM command.	* Change COM command. * Change COM command. * Change COM command. * Change variable.

ERROR CAUSE	ERROR REMEDY
The expression on the right side is of the type "Double-precision real" and the data type (3rd parameter) is not "D" or "R" for write access of the COM command.	* Change COM command.
The expression on the right side is of the type "Integer" and the data type (3rd parameter) is not "B", "W" or "I" for write access of the COM command.	* Change COM command.
For assignment "=", the left variable is of the type "Integer" and the expression is not of the type "Integer" or "NUL", or the left variable is of the type "Real" or "Double-precision real" and the expression is not of the type "Integer", "Real" or "Double-precision real" or "NUL", or the left variable is of the type "Boolean" and the expression is not of the type "Boolean" or "NUL" or the left variable is of the type "String" and the expression is not of the type "String" or "NUL".	* Change assignment.
2176] MISSING	
The closing square bracket is missing in an IC block for an AXP command, a variable address or a parameter list.	* Change NC block.
2177 (MISSING	
The opening bracket is missing for one of the following CPL instructions: CIR, COL, FIL, GMD, GWD, LIN, MWD.	* Change CPL program.
The opening bracket is missing for one of the commands CLOSE, COM, DELETE INP#, OPENR, OPENW, PRN# or REWRITE.	* Change corresponding command.
The opening bracket is missing for a field variable.	* Change field variable.
The opening bracket is missing for one of the following CPL commands: ABS, ACOS, ASIN, ATAN, AXO, AXP, BCD, BIN, CHR, COS, CPOS, EOF, IC, INT, MPOS, PPOS, ROUND, SD, SIN, SQRT, TAN, TC, TD, TXC or TXT.	* Change corresponding CPL command.
2178 BLOCK NON-EXISTENT	
No branch destination is programmed in the G23/G24 block.	* Change NC block.
An NC block does not exist as the branch destination of a G23/G24 block or a GOTO instruction	* Change NC program.

ERROR CAUSE	ERROR REMEDY
2179 LINE NON-EXISTENT A CPL block does not exist as a branch destination of a GOTO instruction.	* Change CPL program.
2180 LABEL NON-EXISTENT A LABEL does not exist as the branch destination of a GOTO instruction.	* Change CPL program.
2181 INVALID FIELD LIMIT The DIM instruction possesses a field limit > 32767 or the field limit is not programmed as an integer constant.	* Change DIM instruction.
The DIM instruction for a character field has a field limit > 127 or < 1.	* Change DIM instruction.
1. SD (2 ...): The 2nd parameter is < 1 or > 3 or more than 2 parameters are programmed.	* Change parameter correspondingly.
2. SD (5...): The 2nd parameter is < 1 or > 3 or the 3rd parameter is < 1 or > 2.	
3. SD (10...): The 2nd parameter is < 1 or > 2 or more than 2 parameters are programmed.	
4. SD (11...): The 2nd parameter is < 1 or > 2 or the 3rd parameter is < 1 or > 2.	
5. SD (12...): The 2nd parameter is < 1 or > 2 or more than 2 parameters are programmed.	
6. CPOS command: The specified axis is an asynchronous axis with external setpoint input.	* Change CPOS command.
2182 INVALID FIELD TYPE The field variable possesses a field variable which is not "Real" or "Integer" in the DIM instruction.	* Change DIM instruction.
2183 INVALID FIELD INDEX A parameter of a field variable or a parameter of one of the following CPL commands is not of the type "Integer" or "Real": AXO, BCD, BIN, CPOS, FXC, IC, MPOS, PPOS, SD, TC or TD.	* Change CPL program.

ERROR CAUSE	ERROR REMEDY
2184 INVALID PARAMETER	
The group number (1st parameter) of an SD function is not permitted.	* Change SD function.
SD(1,...): The second parameter is less than 1 or greater than 32 or the third parameter is less than 1 or greater than 2.	* Change SD(1,...).
SD(3,...): The second parameter is less than 1 or greater than 2 or more than two parameters are programmed.	* Change SD(3,...).
SD(4,...): The second parameter is less than 1 or greater than 2 or more than two parameters are programmed.	* Change SD(4,...).
SD(6,...): The second parameter is less than 1 or greater than 7 or more than two parameters are programmed.	* Change SD(6,...).
SD(8): More than one parameter is programmed.	* Change SD(8,...).
SD(9): More than one parameter is programmed.	* Change SD(9,...).
SD(13): More than one parameter is programmed.	* Change SD(13,...).
AXO, CPOS, MPOS, PPOS: The specified axis is not defined.	* Change axis number.
IC command: The interface bit (1st parameter) is outside the range permitted in the MPP P4010 for write access or is < 0 or > 255 for read access.	* Change IC command.
AXP instruction: The specified axis is not defined.	* Change AXP instruction.
AXP instruction: The specified axis is already programmed in the NC block.	* Change NC block.
A parameter of the commands CIR, COL, FIL, LIN or one of the first two parameters of the DSP command is greater than 2147483647 or less than -2147483648.	* Change CPL program.
CIR command: The fifth parameter "Direction of rotation" is not 0 or 1.	* Change CIR command.
COL command: One of the three possible parameters is exactly 8, exactly 9, less than 0 or greater than 17.	* Change COL command.

ERROR CAUSE	ERROR REMEDY
FIL command: The third parameter "Filling pattern" is less than 0 or greater than 5 or the fourth or possible 5th parameter is exactly 8, exactly 9, less than 0 or greater an 17.	* Change FIL command.
GMD command: The parameter is less than 0 or greater than 4.	* Change GMD command.
GWD command: The first parameter is greater than or equal to the second one or the third parameter is greater than or equal to the fourth one or the first is less than 0 or the third is less than 0 or the second is greater than 511 or the fourth is greater than 359.	* Change GWD command.
MWD command: The first parameter is greater than or equal to the second one or the third parameter is greater than or equal to the fourth one.	* Change MWD command.
The parameter of the CHR\$ command is not of the type "Integer".	* Change CHR\$ command.
The parameter of the CHR\$ command is less than 0 or greater than 255.	* Change CHR\$ command.
The first letter of the third parameter of the COM command is not "B", "D", "I", "R" or "W".	* Change COM command.
Trying to convert a value < 0 or a value > 99999999 for BIN-BCD conversion.	* Only convert values between 0..99999999.
Trying to convert single- or double-precision real values in BIN-BCD or BCD-BIN conversion.	* Convert only integer values with the value range 0..99999999.
Wrong parameters in CPL command TC for table number or compensation selection number or compensation value.	* Check parameters in the CPL command TC in the part program and change if necessary.
Wrong parameters in CPL command FXC for zero shift table or zero shift number or axis number.	* Check parameters in the CPL command FXC in the part program and change if necessary.
Wrong number of parameters for the CPL TD command.	* Change CPL program.

ERROR CAUSE	ERROR REMEDY
The 1st or 2nd parameter is outside the permitted range.	* Permissible range: 1st parameter: Minimum: 1 Maximum: number of variables in the KS table 2nd parameter: Minimum: 1 Maximum: number of groups in the KS table
The parameter of the GMD command is not of the type "Integer".	* Change GMD command.
The four parameters of the GWD command are not of the type "Integer" or fewer than four are programmed.	* Change GWD command.
The four parameters of the MWD command are not of the type "Integer" or "Real" or fewer are programmed.	* Change MWD command.
The three possible parameters of the COL command are not of the type "Integer".	* Change COL command.
The four possible parameters of the LIN command are not of the type "Integer" or "Real".	* Change LIN command.
The six possible parameters of the CIR command are not of the type "Integer" or "Real".	* Change CIR command.
The first two parameters of the FIL command are not of the type "Integer" or "Real" or the third, fourth or possible fifth parameter is not of the type "Integer".	* Change FIL command.
The OPENW command has more than five parameters.	* Change OPENW command.
The OPENR command has more than three parameters.	* Change OPENR command.
The DELETE command has more than two parameters.	* Change DELETE command.
The CLOSE or REWRITE command has more than one parameter.	* Change corresponding command.
The program name of the OPENW command is not of the type "String".	* Change OPENW command.
The file No. or the length in the OPENR or OPENW command is not of the type "Integer".	* Change corresponding command.

ERROR CAUSE	ERROR REMEDY
The DELETE, CLOSE or REWRITE command has more than one parameter.	* Change corresponding command.
One of the first two parameters of the COM command is not of the type "Integer".	* Change COM command.
The third parameter of the COM command is not of the type "String".	* Change COM command.
The third parameter of the COM command is a string constant and does not start with a "B", "D", "I", "R" or "W".	* Change COM command.
The third parameter of the COM command is not a variable and not a string constant.	* Change COM command.
One of the following CPL commands has more than one parameter: AXO, BCD, BIN, CPOS, IC, MPOS or PPOS.	* Change corresponding command.
The SD command has more than 2 parameters.	* Change SD command.
One of the following CPL commands has more than three parameters: TC, TD or FXC.	* Change corresponding command.
The first parameter of one of the following commands is not of the type "Integer": CHR, EOF or TXT.	* Change corresponding parameter.
The first parameter of one of the following commands is not of the type "Integer", "Real" or "Double-precision real": ABS, ACOS, ASIN, ATAN, COS, INT, ROUND, SIN, SQRT or TAN.	* Change corresponding parameter.
The parameter of the EOF command is not a variable and is not an integer constant.	* Change EOF command.
An integer constant in the EOF command does not consist of only one digit.	* Change EOF command.
WAIT instruction: The first parameter is a variable and not equal to the fixed predefined variables ACTIVE or READY at this point.	* Change WAIT instruction.
WAIT instruction: The first parameter is not a variable and does not start with "IC" or "NOT".	* Change WAIT instruction.
WAIT instruction: There is no comma (",") or the bracket is not closed ["] after the IC query or NOT expression in the first parameter.	* Change WAIT instruction.
WAIT instruction: The second or third parameter is not of the type "Integer".	* Change WAIT instruction.

ERROR CAUSE	ERROR REMEDY
WAIT instruction: There is no comma (",") or the bracket is not closed [")"] after the second parameter.	* Change WAIT instruction.
WAIT instruction: The third parameter is not a stand-alone variable.	* Change WAIT instruction.
AXP instruction: The first parameter is not of the type "Integer".	* Change NC block.
AXP instruction: There is no comma (",") after the first parameter.	* Change NC block.
AXP instruction: The second parameter is not of the type "Integer", "Real" or "Double-precision real".	* Change NC block.
AXP instruction: There is something else as well as AXPs in the list enclosed by square brackets, e.g.: [AXP (1,100) ,A].	* Change NC block.
AXP instruction: AXP is programmed in a variable address, e.g.: X[AXP (1,1000)].	* Change NC block.
2185 MULTIPLE SUBROUTINE PROHIBITED	
More than one subroutine call is programmed in the NC block.	* Change NC block.
2186 INVALID PERM-VARIABLE	
A permanent variable was programmed with more than 2 digits; however, only permanent variables from 0..99 are possible.	* Check and correct permanent variables.
The permanent variable does not start with a digit.	* Change permanent variable.
The permanent variable is of the type "Double-precision real", "String" or "Boolean".	* Change type of permanent variable.
The permanent variable is not of the type "Integer" or it is a field variable.	* Change permanent variable.
The number of the permanent variable (first and second digits after the "@") is greater than 50.	* Change permanent variable.
2187 " MISSING	
The final ' ' of a string constant is missing.	* Change string constant.

ERROR CAUSE	ERROR REMEDY
2188 PARAMETER ERROR The directory is not "TTY" and no length is programmed for the OPENW command. A subroutine with more variables than declared in the subroutine is called in the main program of DA mode. A parameter of a current parameter list does not exist, e.g. "P5" (for 5th parameter), in the calling subroutine. No comma (" , ") follows the logical number for the OPENR or OPENW command. No comma (" , ") follows the file No. for the OPENW command.	* Change OPENW command. * Check and correct number of transfer parameters. * Change parameter list or subroutine. * Change corresponding command. * Change OPENW command.
2189 INVALID LABEL A "." is recognized as a label which does not start with a letter.	* Change CPL program.
2190 ERROR IN NC BLOCK If one of the address letters I, J, K, F is missing in G993 of the oscillating axis. The format of the G, M or auxiliary function word (machine parameter P9000) is not permitted. An auxiliary function with transfer-stop is programmed alongside G92. No line feed follows the M2 or M30 word (except for blanks). The NC block has more than 128 characters or it does not end with a line feed at the file end. In "axis coupling", the address programmed is not that of a slave axis (MPP124/2 or 124/4).	* Change part program. * Change corresponding NC word. * Change NC block. * Change NC block. * Change NC block. * Change NC program.

ERROR CAUSE	ERROR REMEDY
Two of the following NC words are programmed in one block: G70, G71, G75, G33 or G92.	* Change NC block.
A G subroutine call is not the last G-word in a block.	* Change NC block.
The M subroutine call is not the last M-word in a block.	* Change NC block.
An auxiliary function subroutine call is not the last corresponding auxiliary function word in a block.	* Change NC block.
There is no address letter before a “[” in the NC block and it is not a parameter list (no subroutine call in the NC block) and it is not an AXP instruction.	* Change NC block.
G491 programmed together with other G codes.	* Program G491 in a single block.
Address “C” is programmed with active G491.	* Avoid “C” address.
G491 is programmed with non-active C-axis.	* Program G491 with active C-axis.
A faulty NC block was discovered during the syntax check for DA operation.	* Change NC block.
Moving through turning centre while machining face end.	* Change NC program.
2191 DIVISION BY 0	
CPL division “/” by zero exists. If CPL division is not involved, then there is an internal software error.	* Check CPL program. * Inform after-sales service.
2193 INVALID TAN	
Tangent calculation of 90 degrees, –90 degrees.	* Calculate only values which are not 90 or –90.
2194 INVALID ASIN ACOS	
For calculation of the arc sin or arc cos of a value outside the value range.	* Observe value range: –1 ... 1
2195 NEGATIVE ROOT	
Root calculation of a negative number. Argument for root calculation is negative.	* Only positive values permitted

ERROR CAUSE	ERROR REMEDY
2196 INVALID REAL VALUE Occurs for the ROUND or INT command if the REAL value to be processed is outside the value range. An overflow has occurred in conversion of a double-precision real number into a single real number. Result of the root calculation is too large. With coordinate transformation switched on, turning a programmed position causes a REAL overflow.	* Observe value range of the REAL value -2147483648 ..2147483647. Check CPL program. * Only positive values permitted. * Change NC program.
2197 PROHIB. WORKING AREA The programmed end point is in the prohibited working area.	* Check program.
2210 ENDDLG MISSING There is a second DLG without the first having been terminated by ENDDLG. The corresponding ENDDLG is missing for a DLG in a subroutine.	* Change CPL program. * Change CPL program.
2211 DLG MISSING There is an ENDDLG without a DLG. There is a PRN command with fourth parameter outside a DLG sequence. An INP command has been recognized outside a DLG sequence.	* Change CPL program. * Change CPL program. * Change CPL program.
2212 WRONG INPUT FORMAT The opening bracket after the INP command is missing. The parameter of the INP command is not a variable and not a field variable. The closing bracket of the INP command is missing.7	* Change INP command. * Change INP command. * Change INP command.

ERROR CAUSE	ERROR REMEDY
2213 WRONG PRINT FORMAT	
GPR command: The first parameter is less than 0 or greater than 511, or the second parameter is less than 0 or greater than 350.	* Change GPR command.
DSP command: A parameter is not one of the type "Integer", "Real", "String" or "Boolean".	* Change DSP command.
PRN command: The first parameter is less than 4 or greater than 42 or the second parameter is less than 1 or greater than 64.	* Change PRN command.
PRN command: Opening bracket missing.	* Change PRN command.
PRN command: The first or second parameter is not of the type "Integer" or "Real".	* Change PRN command.
PRN command: There is no comma after the first or second parameter.	* Change PRN command.
PRN command: The third parameter is not of the type "String".	* Change PRN command.
PRN command: One of the two first parameters is not an integer constant or the third parameter is not a string constant within the DLG sequence.	* Change PRN command.
PRN command: The fourth parameter is not a variable or field variable.	* Change PRN command.
PRN command: No closing bracket follows the third or fourth parameter.	* Change PRN command.
PRN#(0..) command: Two commas are programmed in sequence.	* Change PRN#(0,..) command.

ERROR CAUSE	ERROR REMEDY
PRN#(0...) command: A comma is missing between two parameters.	* Change PRN#(0,...) command.
PRN#(0...) command: Something other than comma, variable, field variable, string constant, the function TXT or the function CHR is programmed in the parameter list.	* Change PRN#(0,...) command.
INP#(0...) command: Two commas are programmed in sequence.	* Change INP#(0,...) command.
INP#(0...) command: A comma is missing between two parameters.	* Change INP#(0,...) command.
INP#(0...) command: A parameter is not of type "Integer", "Real" or "Boolean".	* Change INP#(0,...) command.
INP#(0...) command: Something other than a comma or variable is programmed in the variable list.	* Change INP#(0,...) command.
No comma (",") and no closing bracket [")"] follows after a variable for the INP#(1,) to P#(9,) command.	* Change INP# command.
Two commas (",") are programmed in sequence for the INP#(1,) to INP#(9,) commands or the PRN#(1,) to PRN#(9,) commands.	* Change corresponding command.
A variable as from the second parameter in the INP#(1,) to INP#(9,) command is not of type "String".	* Change INP# command.
Something other than a comma, variable or closing bracket [")"] is programmed as from the second parameter of the INP#(1,) to P#(9,) command.	* Change INP# command.
There is no comma (",") or closing bracket [")"] for the PRN#(1,) to PRN#(9,) command as from the second parameter after a variable, a string constant, a field variable, the TXT\$ command or the CHR\$ command.	* Change PRN# command.

ERROR CAUSE	ERROR REMEDY
2214 WRONG SOFTKEY FORMAT SFK command: Opening bracket missing. SFK command: First parameter is a field variable and it is not of the type "Integer" or it has more than two parameters. SFK command: First parameter is a variable and is not of the type "Integer" or "String". SFK command: First command is not a variable and not a field variable. SFK command: There is no comma after the first parameter. SFK command: No string expression is programmed except for the first parameter. SFK command: More than six parameters are programmed. SFK command: A comma is missing between two parameters. SFK command: A parameter is not of the type "String". SFK command: Something other than a comma, string constant or variable is programmed in the parameter list.	* Change SFK command. * Change SFK command. * Change SFK command. * Change SFK command. * Change SFK command. * Change SFK command. * Change SFK command. * Change SFK command.
2216 PRN INSTRUCTION MISSING An INP command in a DLG sequence does not have an assigned PRN instruction. A PRN command of a DLG sequence (with fourth parameter) does not belong to the active DLG sequence.	* Change CPL program. * Change CPL program.
2217 INPUT INSTRUCTION DOUBLE There is a second INP instruction with the same parameter in the DLG sequence.	* Change or delete an INP instruction.
2219 PRN INVALID A PRN command of a DLG sequence does not belong to the active DLG sequence.	* Change CPL program.
2220 FILE NON-EXISTENT Trying to open a file which does not exist with the OPENR command.	* Change OPENR command.

ERROR CAUSE	ERROR REMEDY
2221 LOG. NUMBER INVALID A logical number is programmed which is less than 1 or greater than 9 for one of the commands OPENR, OPENW, CLOSE or REWRITE. The logical number is not 0 and is less than 1 or greater than 9 for the INP# or PRN# command. No logical number is specified for the OPENR or OPENW command at which a file has already been opened. The first parameter of the OPENR, OPENW, CLOSE or REWRITE command is not of the type "Integer". The first parameter of the INP# or PRN# command does not start with a variable and is not an integer constant. The first parameter of the INP# command is an integer constant and consists of several digits.	* Change corresponding command. * Change corresponding command. * Change CPL program. * Change corresponding command. * Change corresponding command. * Change INP# command.
2222 FILE NOT OPENED Trying to read a file with INP# which is not opened for reading. Trying to rewrite a file by means of REWRITE which is not opened for writing. Trying to write a file or interface with PRN# which is not opened for writing.	* Open file previously with OPENR. * Open file previously with OPENW. * Open previously with OPENW.
2223 FILE END REACHED Trying with INP# to read from a file whose read pointer is already at ETX or at the end of the file.	* Check CPL program.

ERROR CAUSE	ERROR REMEDY
2224 FILE IN USE	
Trying to open a file with OPENR or OPENW which has already been opened by CPL. 2. Trying with OPENW to open an active file.	* Close files, e.g. by control reset. * Check CPL program or active files.
The main program or a subroutine is being used by CPL and therefore cannot be incorporated in the link table.	* Quit CPL program and restart linking.
File cannot be opened by CPL when linking is active.	* Abort linking or wait until linking has finished.
A file with the same program no. has already been opened with OPENW before FIXB.	* Change program no. for OPENW or FIXB.
2225 INTERFACE INVALID	
A TTY string was programmed for OPENW that is not equal to "TTY1" or "TTY2".	* Change OPENW command.
A file number was programmed that is not of the type "Integer" after the "TTY" string for the OPENW command.	* Change OPENW command.
A file no. was programmed for the OPENW command which is not of the type "Integer".	* Change OPENW command.
An interface which is invalid in the COCOM version was addressed in the OPENW command.	* Change OPENW command.
Error outputting monitored ranges (diagnosis control): Invalid interface selected.	* Select other interface.
2226 FILE LENGTH INVALID	
The parameter with the length value is not of the type "Integer" in the OPENW command.	* Change OPENW command.
The length value is less than 130 in the OPENW command.	* Change OPENW command.
Something other than a closing bracket [")"] is programmed as well as the length and file name in the OPENW command.	* Change OPENW command.

ERROR CAUSE	ERROR REMEDY
2227 BLOCK > 128 BYTES	
The program identifier in the OPENR or OPENW command is longer than 12 characters.	* Change OPEN command.
The string to be read into the file by means of INP# is longer than 127 characters.	* Change file to be read in.
Trying to transfer more than 127 characters when writing in a file by means of PRN#.	* Change PRN# command.
INP#(1...9) command: Trying to read data out of a file whose length exceeds 128.	* Correct length or use several variables.
2228 FILE NAME INVALID	
The TTY string in the OPENW command is longer than 4 characters.	* Change OPENW command.
Program name for FIXB is not a string expression.	* Change program name of FIXB.
Trying to open a file in the L directory for writing.	* Change OPENW command.
The file no. in the OPENR or OPENW command is < 1 or > 999999999.	* Change OPEN command.
The file no. in the OPENR or OPENW command is not of the type "Integer".	* Change OPEN command.
A program identifier in the OPENR or OPENW command has other letters apart from the directory letters or more than 9 digits.	* Change OPEN command.
The second parameter of the OPENR or OPENW command is not of the type "Integer" and not of the type "String".	* Change OPEN command.
Trying to open a TTY interface for reading.	* Change OPENR command.
The file no. in the OPENR or OPENW command is not of the type "Integer".	* Change OPEN command.
2241 RANGE EXCEEDED	
The permitted range for the COM command was exceeded.	* Change COM command.

ERROR CAUSE	ERROR REMEDY
2242 UNAUTHORIZED ACCESS The COM command wants read access to a range although this range is not enabled for reading. The COM command wants write access to a range although this range is not enabled for writing. COM command is not permitted since there is no communication range.	* Change COM command. * Change COM command. * Remove COM command.
2243 RANGE NOT PRESENT A range is specified which is less than 0 or greater than 9 in the COM command or the specified range is not enabled for reading or writing.	* Change COM command.
2244 WRONG COM FORMAT The range or address is missing in the COM command.	* Change COM command.
2245 REAL NUMBER OVERFLOW An overflow has occurred in a real calculation.	* Check CPL program. If an NC program, inform after-sales service.
2248 CHARACTER AREA NOT OCCUPIED Trying to assign a value to a component of the character area. However, previous components are not occupied. Trying to assign a value to a component of the character area or read a value from it. However, previous components are not occupied or a component is accessed which is not in the dimensioning limits. A character area variable is still undefined.	* First describe previous components. * First describe previous components or check dimensioning. * Allocate a value to the character area.
2249 BLOCK > 127 BYTES INP#(1...9) command: Trying to write data to a file with a data length > 127 bytes.	* Correct length or use several variables.
2257 INVALID FILE A sequential file has been accessed with the SEEK command. The file specified in FIX is not an image file.	* SEEK is only allowed for a RANDOM file. * Change file no. for FIX.

ERROR CAUSE	ERROR REMEDY
2258 INVALID INTEGER INP# (1...9) command: An excessively high value is assigned to an integer variable. INP# (TTY–SS) command: The integer to be processed is outside the value range.	* Observe value range –999999999 to +999999999 for integers. * Observe value range.
2290 INVALID VALUE The expression on the right side is < –128 or > 127 for bitwise write access by the COM command. The expression on the right side is < –32768 or > 32767 for wordwise write access by the COM command. The end of the specified range is exceeded in wordwise write access by the COM command. Programming G876, G877 makes the upper limit switch value smaller than the lower one.	* Change COM command. * Change COM command. * Change COM command. * Change NC program.
2355 DNC ACTIVE A function which uses the DA buffer is already active.	* Abort the other function.
2389 CNC IN LOCAL MODE DA mode called in LOCAL mode.	* Switch NC to REMOTE.
2391 NO MEMORY ACCESS The ring memory cannot be created in DA mode since the memory is occupied by another function.	* Abort function which is currently using memory.
2395 NC RUNNING A program is already being executed when DA mode is activated.	* Set FEED HOLD, cancel program.
2398 INVALID FILE NO. The file no. for FIXB or FIX is not an integer expression or is not between 1 and 999999999.	* Change file no.

ERROR CAUSE	ERROR REMEDY
2669 PLC TIMING ERROR The dual port to the PLC is accessed in the NC block with G160, external zero shift, and this dual port is blocked by the PLC. Access conflicts occur between the CNC and PLC during reading of external zero shift values.	* Check PLC program. * Inform after-sales service.
3001 RADIUS TOO SMALL The radius for circular interpolation is too small for tangential tool guidance.	* Increase radius in part program.
4695 FUNCTION ALREADY ACTIVE The function is already active. DA mode is already active when activated. Before FIXB, an image recording was started and not completed.	* Wait. * Abort DA mode. * Complete previous image recording with FIXE.
4698 WRONG PROGRAM DA mode cannot be activated since the sent program does not correspond to the requested one.	* Send requested program.
4804 KS ACCESS BLOCKED The PLC blocks access to a KS table (CPL TD command). PLC blocks access to the W file for an inadmissibly long time. (Reading in or output of KS tables).	* Check whether PLC is running. Shorten access to KS table in the PLC program. * Remedy error in the PLC program.
4805 KS DATA CORRUPT Reading-in of KS tables, syntax error in the data part: address-oriented reading-in: no variable (address letter) after "&"; group number not permitted; variable does not exist in the KS table; invalid special character; deletion of groups incorrectly terminated.	* Correct data part of KS table accordingly.
4810 KS-DEF: TEXT LENGTH Reading-in of KS tables, definition part, text too long: text for variable > 14 characters; text for bits of the status word > 18 characters, group text > 11 characters, text for sub-tables (softkeys) > 10 characters.	* Correct definition part of KS table.

ERROR CAUSE	ERROR REMEDY
4811 KS-DEF: NUMBER OF GROUPS Reading-in of KS tables, number of defined groups not permitted, minimum: 1 maximum: 999.	* Correct definition part of KS table.
4812 KS-DEF: ADDRESSES Reading-in of KS tables, definition of addresses (variables): Format for variable not permitted: – type “Word” max. 4 positions – type “Integer” max. 9 positions, max. 7 positions if decimal digits are used. – type “Real” max. 7 positions – type “BCD” max. 8 positions, no decimal digits allowed. Variable defined several times. More than 26 variables defined. No variable defined. Definition of status variables – Status variable not defined as BCD. – Value is not permitted as status of a variable. – Definition of status bits: Bit number not permitted.	* Correct definition part of KS table.
4813 KS-DEF: STATUSTEXT Reading-in of KS tables, definition of sub-tables (softkeys): Status mask not permitted.	* Correct definition part of KS table.
4814 KS-DEF: GROUP TEXT Reading-in of KS tables, definition of group texts: Group number not permitted, minimum: 1 maximum: 999 start group end group	* Correct definition part of KS table.
4815 KS-DEF: SUB-TABLE Reading-in of KS tables, definition of sub-tables (softkeys): Address mask not permitted.	* Correct definition part of KS table.

ERROR CAUSE	ERROR REMEDY
4816 KS-DEF: SYNTAX Reading-in of KS tables, creation of definition part: – Definition part empty. – “\$” is not followed by “[”. – No valid identifier after “[”. – The definition of the address letter is not followed by a comma. – Type of a variable not “W” (word), “I” (integer), “R” (real) or “B” (BCD). – The “,” is missing after definition of a variable type. – The “,” is missing after definition of the format of a variable. – The “,” is missing after definition of the status of a variable. – The “,” is missing after definition of the bit no. for a text of the status variable. – The “,” is missing after definition of the start group for a group text. – The “,” is missing after definition of the end group for a group text. – The “,” is missing after definition of the status mask of a sub-table. – The “,” is missing after definition of the address mask of a sub-table. – The “,” is missing after definition of the address mask of the additional identifier for a sub-table. – The “,” is missing after definition of the parameter mask of the additional identifier for a sub-table.	* Correct definition part of KS table. * Correct definition part of KS table.
4818 KS-DEF: SUB-TABLE ADDITION Reading-in of KS tables, additional definitions for sub-tables – Address mask not permitted. – Parameter mask not permitted.	* Correct definition part of KS table.

ERROR CAUSE	ERROR REMEDY
4825 KS-SYS: LIST MISSING Address-oriented reading-in of KS tables: The list for the search data could not be created. No access to the list possible with the search data.	* NC reset.
5100 M-WORD MISSING A G32 block is programmed without M-word.	* Insert the M-word in the G32 block.
5101 SPEED MISSING A G32 block is programmed without speed.	* Insert the speed (S-word) in the G32 block.
5102 FEEDRATE MISSING A G32 block is programmed without feedrate.	* Insert the feedrate (F-word) in the G32 block.
5103 INVALID M-WORD An invalid M-word is programmed in the G32 block.	* Program a valid M-word (M3, M4, M13, M14).
5104 INVALID PITCH An F/S ratio is programmed in the G32 block which is outside the permitted range of 0.001...1000 mm/rev.	* Change F-word or S-word.
5105 INVALID AXIS A drilling axis is programmed which is not on the spindle servo card. For the axis programmed with G876, G877, no limit switch range is defined in the MP group 6200. G875 programmed with invalid axis address. Thread cutting without compensating chuck (G32) on the face end (G491) with an axis that is not Z.	* Change axis allocation of the servo card. Drilling axis and spindle must be on one servo card. * Change NC program or define range. * Program only the axes entered in MP 8006/51/ and 8006/52/. * G32 on the face end is only possible with the Z axis.

6. SYSTEM ERRORS

A system error has highest priority; correspondingly, the message differs from conventional error display. The error expresses itself in the fact that all actions are aborted, system standby is cancelled and the machine tool is switched off, for example.

A system error message can be caused by a hardware or software error.

The current display is preserved while "System error" is displayed in the prompt line with "Error number". In addition, the softkeys "RESET", "BOOT" and "DISPLAY" are offered.

Example:

```
-----
System Error 260
-----
RESET      |          | BOOT      |          | DISPLAY
```

The following reactions are initiated by pressing the respective key:

DISPLAY

The error text appears on the screen, in some cases with address. The address can be specified to the Bosch after-sales service for the purpose of error diagnosis.

Example:

```
NMI

129C03    1219587

PRESS KEY TO RESTART
```

```
-----
System Error 260
-----
RESET      |          | BOOT      |          | DISPLAY
```

RESET

You generate a "Clear logic" (control power-up). If the error condition should still exist, then "BOOTING" is necessary (refer to Page 6-2).

BOOT

You re-establish the original control configuration (back-up).

The softkey line follows, with various options for loading the operating system.

```
-----
FROM      FROM      DOWNLOAD
EPROM     EEPROM     ...
-----
```

Caution: All data stored in the RAM and not saved in the EEPROM are lost.

A system error message is the most critical error message from the control. The Bosch after-sales service must be informed even if the error can be remedied using the two provided options.

* Software error

If the displayed system error is based on an internal software error (SW error), the following procedure is recommended:

1. Press SK "DISPLAY"; note error number, error message and address (if displayed).
2. Also make a note of the preceding operating steps, programming and other circumstances which led to the current condition.
3. Attempt to restart the system with SK "RESET" (clear logic).

If the system error should occur again immediately, load the operating system back-up with "BOOT". Caution: All data stored in the RAM and not saved in the EEPROM is lost.

4. The responsible Bosch after-sales service must be informed under all circumstances since the error may occur repeatedly.

ERROR CAUSE		ERROR REMEDY
1	CLEAR LOGIC The diagnosis program CLEAR WHOLE MEMORY has been selected.	* Boot or reset.
2	WATCHDOG COUNTER P2 NOT RUNNING P2 not running or P1 running too fast. Processor P2 no longer updates counter in dual port. — Defective or wrong P2—SW. — RAM of processor P2 faulty, processor P2 faulty, HW of P2 faulty	* Check MP 9902. * Check or replace SW. * Replace CP subassembly.
6	KS-DEF. PART DESTROYED The definition part of a KS table which is currently being accessed has been destroyed. The PLC has overwritten the definition part of the KS table. KS table not in permitted area.	* Reset; Attempt to output all KS tables. The table where the system error occurs again has a fault in the definition part. If the system error does not occur, there is probably a fault in the NC operating system. If the error occurs again, clear all KS tables, create again and read in the saved data after a reset. * Check KS tables; inform after-sales service.
14	FILE TRANSFER ERROR Termination of output logic: — Mail sent with invalid parameters. — File length < 0. — Invalid file group for output of a file. — File can no longer be found after control reset. — Output buffer can no longer be found after control reset.	* Inform after-sales service.

ERROR CAUSE	ERROR REMEDY
Termination of read-in logic: – Mail sent with invalid parameters. – Trying to close a file, but this has a length < 0. – Invalid mode in TEST ISO. – Invalid mode in READ IN. – Invalid mode in DEIN TRANSFER – Inadmissible jump out of an application. – Start data pointer points to NIL. – Parameter error and utility error for FILE NO. STATUS in TEST FILE.	* Inform after-sales service.
27 ACTIVE TABLE MISSING Active compensation table does not exist. Active zero shift table does not exist.	* Inform after-sales service.
99 IS MAIN BLOCK SEARCH WANTED? An incorrect task number is specified internally when the CPL flag is reset.	* Reset.
125 INADMISSIBLE COMPENSATION VALUE System error occurs with active cutter radius compensation if the logic for the intersection point calculation generates an internal error message which is unknown in the error processing module.	* Inform after-sales service.
193 INVALID TANGENT Internal computing error.	* Inform after-sales service (refer to Page 6–2).
195 NEGATIVE ROOT Internal computing error.	* Inform after-sales service (refer to Page 6–2).
196 INVALID REAL-VALUE Internal computing error.	* Inform after-sales service (refer to Page 6–2).
202 TIMING ERROR IN WAITFORRXR Error during waiting for WAITFORRXR.	* Check connection between panel and CP; replace CP subassembly.
203 NO INTERROGATION Error during waiting for WAITFORNCRT.	* Check connection between panel and CP; replace CP subassembly.
231 TIMING ERROR IN PLSINI Error during waiting for WAITFORRXR.	* Check connection between panel and CP; replace CP subassembly.

ERROR CAUSE		ERROR REMEDY
232 RAM-ECHO IN PLSINI Written-in values are read out incorrectly during the internal test. HW error.		* Replace CP subassembly. * Check panel hardware.
234 NO RXR-BIT FOR PANEL INTERRUPT Error in the panel software or hardware error at the party line interface.		* Check Interface. * Replace CP subassembly
253 WRONG INTERRUPT PCB Interrupt controller does not send permitted interrupt VECTOR 4.		* Inform after-sales service * Replace CP card.
256 COUPLING BOARD RESET The coupling board and MEM board do not match with respect to software. The memory area on the coupling board, PC600 word coupling, has been cleared. The NC does not have any information about how the data is to be transferred to the PLC.		* Load backup. * Reload KS definition tables.
260 NMI Signal at the NMI input of the processor (power fail) or pushbutton on the CP subassembly pressed. SW error.		* NMI pushbutton on CP switch defective. * HW error on the CP subassembly; replace CP board. * All pins of the processor module correctly inserted? * Inform after-sales service (refer to Page 6–2).
261 ABORT Signal at ABORT-PIN (CP) of the processor (HW error). SW error.		* All pins inserted correctly? * Replace CP board. * Inform after-sales service (refer to Page 6–2).
262 FPU-TRAP Internal error during a computing operation; invalid memory access.		* Inform after-sales service (refer to Page 6–2).
263 ILLEGAL OPERAND Inadmissible instruction in the program code.		* Inform after-sales service (refer to Page 6–2).
264 SVC-TRAP SVC with invalid parameter in R0. SW error; program overwritten; prohibited monitor call.		* Inform after-sales service (refer to Page 6–2).
265 DIV BY ZERO Inadmissible division by 0. Certain states cannot be handled by SW.		* Inform after-sales service (refer to Page 6–2).

ERROR CAUSE	ERROR REMEDY
266 FLAG-TRAP FLAG command in the program. SW error; program overwritten. Overflow during a com- puter operation.	* Inform after-sales service (refer to Page 6–2).
267 BREAKPOINT REACHED BPT command in the program. Program overwritten. Control SW destroyed.	* Replace EPROM modules. * Replace MEM board. * Inform after-sales service (refer to Page 6–2).
268 TRACE TRAP Interruption for debugger.	* Inform after-sales service (refer to Page 6–2).
269 UNDEFINED OPCODE Program overwritten. SW error; control SW destroyed.	* Replace EPROM modules. * Replace SW. * Inform after-sales service (refer to Page 6–2).
270 ERROR: DUAL-PORT-TEST Error during control power-up; HW error.	* Switch control off and on. If an error message still occurs, replace CP board.
271 ERROR: PARTY-LINE TEST Error during control power-up.	* Switch NC and panel off and on. * Interface on CP (party-line) faulty: replace CP board. * Interface on panel control faulty: replace panel control. * Panel self-test with shorting plug at X12. * Inform after-sales service (refer to Page 6–2).
272 ERROR: RAM-TEST Error during control power-up.	* Switch control off and on. * Replace CP board. * Inform after-sales service (refer to Page 6–2).
273 ERROR: RE LOAD TEXT Error on cassette to be reloaded.	* Inform after-sales service (refer to Page 6–2).
274 SVC NO. 6 Control SW is defective.	* Replace EPROM modules. * Replace SW. * Inform after-sales service (refer to Page 6–2).
275 HEAP OVERFLOW Heap over low (RAM), graphic memory el- ements exceeded. Too many graphic data; graphic memory full. SW defective.	* Inform after-sales service (refer to Page 6–2).

ERROR CAUSE		ERROR REMEDY
276 SQRT OF NEG. ARGUMENT	Call of root function with negative operands.	* Possibly error in CPL. * Inform after-sales service (refer to Page 6–2).
277 BACKUP VERSION ACTIVE	Error message after BACKUP loading. BACKUP loading is automatically performed after software loss.	* “Reloading” – with data from EEPROM or EPROM – is possible after input of the correct code. * Replace CP, memory or W-coupler. * Inform after-sales service (refer to Page 6–2).
280 ERROR: P2 INIT	Faulty initialization of P2–SW.	* HW reset; inform after-sales service if necessary.
281 TIMEOUT BY WAIT FOR P2	Processor P2 not running at system start. Wrong P2 software version.	* HW reset; inform after-sales service if necessary.
282 CRITICAL SECTION NOT ENDED	It was not possible to end an internal program section after the control was switched on.	* Inform after-sales service (refer to Page 6–2).
615 EPROM ERROR	P1–EPROM error.	* Inform after-sales service (refer to Page 6–2).
621 CODE RANGE 4 NOT AVAILABLE	A routine was called from code range 4 on MEM4; however, this code range does not exist.	* Inform after-sales service (refer to Page 6–2).
703 MACRO ADDRESS NOT PRESENT FOR RESET	Software error during output of auxiliary function. Software error during processing of an F-address.	* Inform after-sales service (refer to Page 6–2).
704 MSG NOT PRESENT FOR OUTPUT	SW error during processing of the active block.	* Inform after-sales service (refer to Page 6–2).
708 INCORRECT USE OF DO AGAIN FLAG	SW error during processing of G70 or G71.	* Inform after-sales service (refer to Page 6–2).
739 ERR BLOCKCON	SW structure error.	* Inform after-sales service (refer to Page 6–2).

ERROR CAUSE	ERROR REMEDY
798 WRONG MODE IN READ System error during reading-in and output; SW error: – Incorrect mail. – Utility occupied or parameter error. – Utility error. – Output list no longer exists. – Inadmissible readr-task-mode. – File does not exist. – KS table to be output does not have a sub table. – No output of KS table is possible since KS Tab-Editor active.	* Inform after-sales service (refer to Page 6–2).
800 SOFTKEY PARAMETER < 0 Incorrect mode for calling the input utility; SW error.	* Inform after-sales service (refer to Page 6–2).
804 LSEC–FILE DEFECTIVE 1. An invalid address is programmed in the files L101–L108. Program address. 2. A file number of the LSEC files does not have the prescribed value. 3. An error has occurred during transfer of the LSEC file to the servo loop processor D25, which can also occur during the LSEC search.	* Valid addresses: X, Y, U, V, W, A, B, C, R, P, N, S. * Number of LSEC compensation file: "Axis number +100". * Check the LSEC file with the diagnostic program D25.
805 ERROR IN LSEC FILE PROCESSING An inadmissible address was recognized in one of the compensation files L101–108 during the LSEC search. The number of a file read in the LSEC search does not have the axis number +100.	* Permitted addresses: X, Y, U, V, W, A, B, C, R, P, N, S. * The No. of the LSEC compensation files must have the axis number +100. Read in LSEC file again and transfer.
810 FORMAT NOT AGREED Negative SK No. for call of softkey display utility; SW error.	* Inform after-sales service {refer to Page 6–2}.
813 FILE GROUP NOT DEFINED (OPEN AND CLOSE) SW error. Incorrect file group for reading in files. Wrong file group for interrogation of the memory start.	* Inform after-sales service (refer to Page 6–2). * Inform after-sales service. * Inform after-sales service.

ERROR CAUSE	ERROR REMEDY
814 PROGRAM MEMORY DESTROYED! SW error; Control has established an error after closing a part program. The created programs no longer exist. The memories for the P, S, V, K, L, G, W, D files have been destroyed. Error addresses: 1: memory on the MEM or CP/MEM4. 2, 3: memory on the WC card.	* Inform after-sales service (refer to Page 6–2). * Inform after-sales service. * Inform after-sales service.
815 POINTER IN THE LIST OR TREE MANAGEMENT DESTROYED The control internally monitors whether the pointers for the selected programs are within the part or system memory for each access to the part memory and system data. An error message is issued if a deviation is established.	* Load backup; inform after-sales service if necessary.
816 ERROR IN USER HEAP Internal error. Control provides defined data area for operation. If this area is exceeded or irregularities are established, an error message is issued. CAUTION: Data in USER HEAP are destroyed!	* Inform after-sales service. Reset error and USER HEAP with Reset.
817 LSV2 STATUS DATA FAULT Internal SW error.	* Inform after-sales service (refer to Page 6–2).
818 INVALID POINTER WHEN CLOSING A FILE Internal SW error. An error was detected when closing a file.	* Inform after-sales service (refer to Page 6–2).
851 COPY FILE NOT PRESENT A file to be copied does not exist in the part program memory.	* Inform after-sales service (refer to Page 6–2).
950 STACK SUBRANGE EXHAUSTED The last active task has exceeded its permitted stack range.	* Inform after-sales service (refer to Page 6–2).
990 ILLEGAL MEMORY ACCESS Internal SW error.	* Inform after-sales service (refer to Page 6–2).
991 TASK MISSING IN NOTQ Software contains errors.	* Inform after-sales service (refer to Page 6–2).

ERROR CAUSE		ERROR REMEDY
992 TASK STILL ACTIVE	Task cannot be transferred to another queue since it does not exist in the system.	* Inform after-sales service (refer to Page 6–2).
993 TASK END WITHOUT STOP	SW error; task is ended without calling the internal function “Stop”.	* Inform after-sales service (refer to Page 6–2).
994 WRONG USE OF OPTIONS	Incorrectly used branch-out in the software or branch-out with error.	* Inform after-sales service (refer to Page 6–2).
995 TASK TRANSFER IMPOSSIBLE	Internal SW error.	* Inform after-sales service (refer to Page 6–2).
998 ILL. PARAM IN UTILITY	Defective software.	* Inform after-sales service (refer to Page 6–2).
	System error occurs: 1. During copying of memory areas if a negative value is transferred as the number of bytes to be copied. 2. During conversion of integer or real numbers from ASCII format to binary format if the end of the number string is not recognized.	
	Internal data error, wrong transfer parameters. Utility called with inadmissible parameters.	* HW reset; inform after-sales service.
999 STACK OVERFLOW	No stack area available for new task initialization	* Inform after-sales service (refer to Page 6–2).
2014 FILE TRANSFER ERROR	System error on the PLC card (read-in logic).	* Replace SCP board; inform after-sales service if necessary.
4641 DNC HARDWARE ERROR	Receive register of a UART on the PLC card is blocked.	* Check interface parameters; RESET. Replace SCP board if the error message is still present.
2672 TSU: ILLEGAL CHANNEL NUMBER	RAM is faulty.	* Replace MEM card.
	SW does not initialize the whole RAM area.	* HW reset/clear logic/control reset.

7. WARNINGS

Warnings are error messages with the lowest priority. They indicate that an error with a higher priority may occur if you continue working and ignore the message, for example.

System warnings are shown as system errors if the test board is connected. It may be possible to achieve a controlled condition again by "Control reset" or "Clear logic". Inform the Bosch after-sales service under all circumstances.

ERROR CAUSE		ERROR REMEDY
66 PROGRAM NOT PRESENT		
The program specified as a trigger condition for an axis oscilloscope recording has not been found.	*	Perform recording with correct trigger condition.
Contour display is to trigger on a non-existent program.	*	Trigger on existing program.
A CPL program is started in machine, data I/O, diagnosis or compensation mode which does not exist.	*	Create corresponding CPL program.
Trying to activate a diagnosis program which is not available.	*	Enter correct program number or provide or read in required diagnosis program.
External program selection; program which is to be selected does not exist.	*	Select a different program.
67 INPUT ERROR		
Invalid values programmed; offset values invalid.	*	Change offset input.
Programmable closed-loop gain (CLG) values invalid.	*	Change CLG values.
72 INCORRECT INPUT		
Trying to traverse a Hirth axis with 1, 10,100 incr. by inching.	*	Inching with a Hirth axis possible only with continuous jog or with the increments set via MPP.
73 INPUT INCORRECT		
Subroutine call is entered in the switching function.	*	Check whether the entered auxiliary function initiates a subroutine call.
78 INVALID FORMAT		
Invalid F values or auxiliary functions were entered in the switching function.	*	Enter new F values or auxiliary functions.

ERROR CAUSE	ERROR REMEDY
79 UPGRADE LEVEL INACTIVE LSV2 is applied, no EPROM memory on the memory board. If the multi-mode at the multi-interface is greater than 26 or less than zero and no application upgrade level is applied for this mode. The NC COMMAND from the PLC is not specified. Optional NC command was recognized. However, the option does not exist. Disturbance compensation: The PLC assigns the task of transferring values to the servo-processor although the upgrade level is not applied.	* Install suitable memory board. * Set multi-mode to permitted value. * Check PLC program. * Check option [20]. * Change PLC program or apply upgrade level.
100 PROGRAM IN CONTROL External program selection when no "Feed hold" is pending.	* Operate "Feed hold".
104 MEMORY FULL A file cannot be created owing to insufficient memory. There is not sufficient free memory for creation of a list for output of main program and subroutine. Free part program memory too small for diagnosis file.	* Clear files which are not required and perform reset. Provide memory space by control reset or by deleting files and control reset. * Create space in part program memory.
There is no longer sufficient free memory in the system to create the data file L402 for the diagnosis program D9. Not enough memory to create list for outputting several files. The free part program memory is not sufficient to create a recording file for the logic analyzer. Free part program memory too small to create the diagnosis program.	* Try to free up memory by deleting files and/or control reset.
105 ERROR IN FILE STRUCTURE Warning during reading-in: PIC file without PE termination.	* Enter PE termination in the PIC file.

ERROR CAUSE	ERROR REMEDY
113 NO REF. POINTS The reference point or the reference point group signal (see MP 1510 reference point required) is missing for a programmed axis when the program is selected. External program selection if not all reference points are present.	* Travel to reference points. * Travel to reference points.
125 BLOCK NON-EXISTENT The block number specified as a trigger condition for an axis oscilloscope recording was not found. Contour display is to trigger on non-existent block in the specified program. Entered block number for block search does not exist in this file.	* Perform recording with correct trigger condition. * Trigger on existing block. * Enter new (existing) block number.
134 TABLE IS ACTIVE A table which is to be deleted is active.	* Cancel table; start deletion once more.
146 PROGRAM ACTIVE A program which is to be deleted is active. Trying to traverse with the handwheel while a program is active. It is wished to traverse an axis by inching or by means of the handwheel while an NC status is running. A CPL program is started in machine, data I/O, diagnosis or compensation mode which is currently selected via NC or manual input mode. Trying to reference machine while program is active. If program is started and interface signal "Feed hold" is pending and INCHING MODE or referencing or axis zeroing is activated. The selected program is active.	* Cancel program; start deletion once more. * Cancel CPL program. * Cancel program. * Cancel program and execute desired function; allow program to run and then execute function. * Cancel the active program.
149 I/O DEVICE: TIMEOUT Error during reading-in: Data do not arrive. Connection is faulty.	* Check interfaces, device connection and device setting.
165 ADDR INVALID "S" is not an auxiliary function. However, an "S" was entered as an auxiliary function.	* "S" must not be programmed as a switching function.

ERROR CAUSE	ERROR REMEDY
187 FILE HAS DFS IDENTIFIER Warning during reading-in: A file number was specified for the file to be read in but this file has a DFS identifier. This has priority. The DFS or FSI identifier stored in the file does not agree with the file No. (reading-in). In the LVS2 file read-in function, the number in the DFS identifier does not match the number in the request telegram. The file is created with the number of the DFS identifier.	* Read in file without DFS identifier. * The file is stored under the number contained in the DFS or FSI identifier. * Check the DFS identifier.
197 F NO. PREALLOCATED An F value cannot be entered later if the most recently programmed F value is 0.	* Program an F value not equal to 0 in the part program.
212 WRONG DIAG. VERSION Tried to start a diagnosis program whose version number does not match the existing operating system.	* Provide desired diagnosis program with the corresponding version number.
217 ERROR IN DIAGN. PRG Tried to activate an incorrectly created diagnosis program.	* Provide correctly produced diagnosis program.
223 CLEAR RESIDUAL DISTANCE G41/42 "Clear residual distance" is initiated with active cutter radius compensation. The contour may be destroyed if work is continued subsequently.	* Restart part program.
243 WRONG CODE NUMBER Wrong code number entered.	* Enter correct code number.
323 EDIT ENABLE MISSING Warning during reading-in: CLM command is not executed since edit enable not present.	* Provide edit enable and repeat reading-in.
332 MAILBOX FULL Output faulty. Read-in logic faulty.	* Control reset
333 PANEL NOT SUITABLE A panel with a color graphics version of less than 2 or a monochrome panel is connected to the control.	* Connect suitable panel.

ERROR CAUSE	ERROR REMEDY
336 LINKING ABORTED Background linking was aborted. While processing via DNC interface, a block received a subroutine call without a link table. In MANUAL INPUT a block was executed that contained a subroutine call without a link table.	* Restart linking.
346 TOO MANY AXIS MOVEMENTS More than 4 axis movements were performed when traveling away from the breakpoint.	* Execute a maximum of 4 axis movements.
355 LINK ERROR An error was established during background linking.	* Remedy error in program and restart linking.
372 LSEC LOGIC The value for the grid increment in an LSEC compensation file was not identified with "S". The table entry for the reference point was found more than once for a traversing direction in an LSEC compensation file. One of the entries for the reference point is missing in one of the LSEC files. The axis position values for the reference point do not agree in the different traverse directions in one of the LSEC files. The minimum number of 2 traverse directions does not exist in an LSEC compensation file.	* The value for the grid increments must be identified with the letter "S". * Only one table entry is permitted for the reference point per traverse direction. * Two entries for the reference point must be contained in each file. * The axis position values of the two entries for the reference point must agree. * There must be at least two table entries per traverse direction.
373 LSEC SYNTAX An axis address specified in the LSEC file does not agree with the MP 102 assigned to the axis. One of the compensation values in the LSEC file is identified neither with "P" nor "N".	* The axis addresses must agree with the corresponding MP. * The compensation values must be identified either by "P" or "N".
374 LSEC GRID An axis position value in an LSEC compensation file is not in line with the prescribed grid increments. The axis position values of the first and last table entry in an LSEC compensation file do not agree.	* All axis position values must lie on a grid defined by the step increment. * There must be a corresponding entry for the opposite traverse direction at the end of the file for the first entry in the table.

ERROR CAUSE	ERROR REMEDY
375 LSEC OVERFLOW It was not possible to perform conversion of an axis position value from an LSEC compensation file correctly. It was not possible to perform conversion of a compensation value from an LSEC compensation file correctly. It was not possible to perform conversion of the value for the grid increment from an LSEC compensation file correctly. The maximum internal LSEC compensation table size of 8000 values defined by grid increments and maximum and minimum axis position values has been exceeded.	* The axis position must be specified with a max. ten-digit number. * Specify the compensation values by a whole number which is not above 32767 and not below -32767. * Specify the value for the grid increment by a max. ten-digit number which does not exceed 2,100,000,000 and is not below 1. * Adapt the grid increments and maximum and minimum axis position values correspondingly.
377 NO CONTROL RESET POSSIBLE External control reset actuated while a program is running. External control reset is actuated and D4 is active. External control reset is actuated and D8 is active. External control reset is actuated and D9 is active.	* Operate "Feed hold" before control reset. * Cancel diagnostic program before control reset. * Cancel diagnostic program before control reset. * Cancel diagnostic program before control reset.
386 LIMIT SWITCH 1st AXIS 387 LIMIT SWITCH 2nd AXIS 388 LIMIT SWITCH 3rd AXIS 389 LIMIT SWITCH 4th AXIS 390 LIMIT SWITCH 5th AXIS 391 LIMIT SWITCH 6th AXIS 392 LIMIT SWITCH 7th AXIS 393 LIMIT SWITCH 8th AXIS A value was specified for synchronous axes which is outside the limit switch range. An axis is traveled to the limit switch with the handwheel or in inching mode.	* Change part program or interface specification. * Travel axis from limit position.

ERROR CAUSE	ERROR REMEDY
399 FILE PROTECTED Warning during reading-in: CLM command is not executed since the file to be deleted is delete-protected. Trying to delete a protected file with DELETE ALL PROGRAMS. The delete or write protection is missing from the selected program.	* Cancel delete protection and continue reading in. * Change file protection, call deletion once more. * Change protection type for the program.
400 INVALID DISTURBANCE TABLE Disturbance compensation: Error in definition of the KS table: Number of variables < number of axes defined in the MPP. Variable not of the type "Word". Structure of the table destroyed by PLC.	* Define more variables in the table. * Define variable as word. * Change PLC program.
401 NO DISTURBANCE TABLE Disturbance compensation: The upgrade level is applied, but the corresponding KS table does not exist.	* Read in KS table.
592 WRONG TOOL TYPE Reading-in of a K table: A defined subdivision of the table into turning and milling compensation groups is performed via MP 7510. The read-in table does not correspond with this MP. However, the table is read in nevertheless.	* Edit the read-in table and correct allocation of the compensation groups to turning and milling tools.
798 INVALID PC OPERAND An invalid PC operand was programmed.	* Check the PLC program; correct operand.
799 INVALID PC COMMAND An invalid PC command was programmed.	* Check PLC program; correct command.
806 EEPROM WRITE-PROTECTED PLC program cannot be programmed in the EEPROM or is aborted with error.	* Switch over switch for write protection on the PLC board. * Replace EEPROM, repeat programming. * Inform after-sales service.
815 PLC BATTERY EMPTY One or more batteries in the word-coupled PC600 is exhausted. An LED on the respective board (NT, RAM or ZE) indicates the status.	* Replace battery(ies); perform control reset.

ERROR CAUSE	ERROR REMEDY
832 PC PROGRAM TOO LARGE The compiled PLC program does not fit in the part program memory.	* Repeat operation after control reset.
1079 CHANGE TO MDI A CPL program which runs under MDI starts a graphic output or starts a dialog and the NC is not in manual input mode. It is an additional requirement for the CPL command SFK that the user is in the basic manual input screen.	* Change to "Manual input" basic screen.
1080 LINK TABLE NOT PRESENT A program is called which has not yet been linked while direct automatic execution is active.	* Linking takes place automatically.
1120 E-DIR. INCOMPLETE There is not sufficient free memory for management of EPROM files.	* Release memory space by clearing file; perform CLEAR LOGIC.
1131 FILE IN LIBRARY Warning during reading-in: File exists in library. The selected program exists in the EPROM library.	* Block file in the library. * Use other program no. for editing.
1168 COMM. STORE USED The communication memory channel for jobs from the NC to the PLC is already occupied by other functions.	* Clear logic.
1169 PLC-FM TIMEOUT The PLC module required for diagnosis is – not present – defective The PLC does not run.	* Incorporate module in PLC program. * Inform after-sales service. * Switch on the PLC.
1170 PLC-FM INCORRECT The PLC diagnosis module acknowledges the job but returns unidentifiable data.	* Inform the after-sales service.
1188 OLD LANGUAGE VERSION The customer EPROM area does not contain the latest texts.	* Program the customer EPROM area with current texts.

ERROR CAUSE	ERROR REMEDY
1189 CUSTOMER EPROM TOO SMALL The customer EPROM area is larger than the applied EPROM area.	* Remove customer EPROM or apply larger area.
1190 CHECKSUM CUSTOMER EPROM A checksum error has occurred during the internal check in the customer EPROM area.	* Replace customer EPROM and reprogram.
1191 CUSTOMER EPROM IN WRONG PLACE Customer EPROM is in wrong location.	* Install customer EPROM in correct location or program the content for the correct location. The location is shown in "GBA diagnosis, other functions, customer area".
1216 FEED OVERRIDE TABLE MISSING Feed override is applied and no feed override table exists.	* Read in feed override table.
1220 TEST ACTIVE "Program test" has been activated after program selection. It is then attempted to start simulation.	* Cancel program test and restart simulation.
1317 MEMORY OVERFLOW The part program memory is full.	* Control reset; clear unnecessary part programs.
1322 OPTIMIZATION BUFFER FULL The reserved area (1 kB) is used for new NC program inputs during program optimization.	* Control reset. Create memory space (delete programs?). Cancel the program awaiting optimization and select again for program optimization.
1390 SIMULATION MEMORY FULL The simulation memory, via which data relative to simulation are forwarded to the output task, is full.	* Increase the size of the simulation memory.
1409 WEAR The limit value is exceeded when editing the parameters DX, DZ and DL, DR (refer to ; MPP P7505).	* Change entry in MPP or possibly remeasure tool.
1470 NO ALT. TOOL No alternative tool whose service life has not expired was found for the programmed tool.	* Check and correct alternative tool combination in the tool editor.

ERROR CAUSE	ERROR REMEDY
1480 ALT. COMPENSATION GROUP INVALID Error during search for an alternative tool. 1. Wrong definition of alternative tool combination (via CPL?). 2. Endless chain in the alternative tool logic. 3. SW error.	* Check and correct the alternative tool combination in the tool editor or check the program for incorrect TC/TD commands. * Control reset and program T again in switching function.
1505 TOOL CHANGE ACKNOWLEDGEMENT The NC must wait for the PLC signal for too long during the tool change.	* Check PLC program for tool change.
1541 CUTTER RADIUS NOT MONITORED The warning is output if 1. the barriers are active and a milling tool is loaded. 2. A milling tool has been loaded and the barriers have been switched from inactive to active.	* In order to reduce the risk of collision with milling tools, the barriers must be increased by the cutter radius. The cutting edge tip of the measuring tool must be the cutter radius away from the barrier for barrier measurement.
1542 BLOCK SEARCH NOT ALLOWED Block search cannot be started while G491 is active.	* Activate G490
1879 PROGRAM ACTIVE A synchronous axis was started by means of external setpoint input, although the NC was not in the condition "NC WAIT". The axes cannot be started for "check claws", since the program is still active.	* Do not enter external setpoint until "NC WAIT" is present. * Cancel program and select new program.
1983 T-TABLE RENAMED A read-in T-table already exists in the technology file.	* Rename table after reading-in.
1984 TABLE NOT ACCEPTED No suitable number is free for renaming of read-in T-tables.	* Rename corresponding tables and repeat reading-in.
1987 PROFILE NON-EXISTENT Tried to start a non-existent profile with G892 or G894.	* Create profile previously with SD function.

ERROR CAUSE	ERROR REMEDY
2003 AXIS RUNNING	
A new value was specified for a synchronous axis whose interpolator is still active.	* Wait with input until the travel command has disappeared or the axis is in position.
NC start with running synchronous axis.	* Stop axes before NC start.
Tried to initiate the function "Clear residual distance" or "Clear logic" or "Control reset" by means of softkey operation in the menu "Intervention function" while a travel command for an axis was pending.	* Function can be performed optionally. When it is, the travel commands are cancelled.
Trying to traverse with the handwheel or in inching mode while the axis is running.	* Stop axes.
Oscillating axis is programmed and the interpolator is still active.	* Stop axis previously or change part program.
Trying to reference machine although not all axes are stationary.	* Stop axes. not all axes are stationary.
Spindle axis without gear speed feedback.	* Signal gear speed back to the NC via the PLC.
2008 FILE NON-EXISTENT	
Warning during reading-in: CLM command is not executed since the file to be deleted does not exist.	* Command unnecessary.
2010 FILE ALREADY EXISTS	
Warning during reading-in: File already exists.	* Delete existing file and repeat reading-in.
2011 FILE IS ACTIVE	
Warning during deletion: CLM command is not executed since the file to be deleted is active.	* Cancel active file and repeat reading-in.
2095 ERROR IN ECODE	
Warning during reading-in: E-code error in a P, L, Spil file.	* Correct E-code value. Read-in file contains corrupt E-codes.
2115 SUBROUTINE CALL MISSING	
Go block/re-entry was selected. A subroutine is specified in MPP P4014. There is no subroutine call or tool change call in the main program.	* Program call in main program or set MP P1014 to "0".

ERROR CAUSE	ERROR REMEDY
2141 DSP FORMAT WRONG The size of an integer format (number of gates) is smaller than the number of positions of the value to be output. An integer format contains a decimal point. The size of a real format (number of gates before the decimal point) is smaller than the number of positions of the value to be output. DSP command attempts to write in line 43. An integer format is greater than 9 gates (#). A real format is greater than 9 gates before the decimal point and greater than 9 gates after the decimal point. The line value is less than 4 or greater than 42 or the column value is < 1 or > 64. The opening or closing bracket of the DSP command is missing. The first or second parameter (line, column) has a result other than "Integer" or "Real". The comma is missing after the first or second parameter.	* Change CPL program. * Change CPL program. * Change CPL program.
2142 NO M2 OR M30 M2 or M30 is missing for reading-in.	* Change part program.
2143 PLEASE ENABLE MEMORY The OPENW, REWRITE or PRN# command requires access to the memory and the latter is busy with a different application (e.g.: editor). The memory is occupied during creation of a list for output of the main program and sub-routine. A link process requires memory space. However, if this is occupied, this warning appears for approx. 1 minute.	* Free up memory. * Execute control reset, repeat function. * Release memory; e.g. by exiting the editor.
2144 MEMORY IS OCCUPIED The part program memory is not released within 1 min. after started simulation.	* Release part program memory and restart simulation.

ERROR CAUSE	ERROR REMEDY
2150 INSUFFICIENT MEMORY The part program memory is too small for creation of the simulation file.	* Increase the free part program memory (clear no longer required programs and/or reset).
2156 P NOT IN CALLCHAIN A subroutine was selected for the program test which is not contained in the current call structure.	* Select other subroutine.
2208 TABLE NOT CREATED The program number of a subroutine is greater than 10 characters when creating a list for output of main programs and subroutines. A subroutine does not exist for output of main programs and subroutines.	* Correct error in main program. * Create subroutine.
2214 WRONG SOFTKEY FORMAT SFK command: A text without "&" is longer than 10 characters or the text before "&" or after "&" is longer than 10 characters.	* Change texts in SFK command.
2218 CHANGE TO AUTO A CPL program which runs under "Automatic" starts a graphic output or a dialog and if the NC is not in the basic operating mode "Automatic". For the CPL command SFK, it is an additional requirement that the user must be in the basic screen for "Automatic". CPL wants to use graphics, but another application is already using it (e.g. simulation).	* Change to the basic operating mode "Automatic". * Stop graphic application with operating mode change and change to "Automatic".
2230 GRAPHICS ALREADY ACTIVE A CPL program is started in machine, data I/O, diagnosis or compensation mode and a CPL graphic program is already active in NC or manual input mode. If the graphics is already occupied by an other function which performs graphic outputs (e.g. CPL) upon entry into the NC simulation block, then this warning is output.	* First terminate CPL graphic program in NC or manual input mode. * Cancel one of the two functions.

ERROR CAUSE	ERROR REMEDY
2246 INVALID COMPONENT LENGTH Different component length in a random file which has been opened for reading. OPENR-OPENW command: The component length is not in the permitted value range.	* Create file anew or correct. * Change to value range 1...127.
2247 INVALID COMPONENT SEEK command: Specified component does not exist.	* Correct SEEK command.
2275 HEAP OVERFLOW Warning during reading-in: No E-code error display possible since insufficient system memory available.	* Initiate "Clear logic".
2281 TIMEOUT WHILE WAITING FOR P2 Disturbance compensation: The PLC forwards the jobs for value transfer to the servo-processor faster than the latter can accept them.	* Change PLC program. Only processing times of 500 ms are guaranteed.
2290 INVALID VALUE The programmed F-value is too large or negative.	* Repeat input with a new F-value.
2310 INTERNAL Error during relocation of a log book (e.g. after reset).	
2357 STARTED BY CENTRAL CONTROL UNIT ONLY Tried to start the resident diagnosis program D26 "Remote diagnosis". Tried to start a reloaded diagnosis program D26.	* The resident diagnosis D26 is reserved for remote diagnosis and can be started only from the remote diagnosis central control unit. * The program number D26 is reserved for remote diagnosis. The program can be started only from the remote diagnosis central control unit.
2358 NO DIAG. START POSSIBLE Tried to start a diagnosis program while the diagnosis program D26 (remote diagnosis) was active.	* Remote diagnosis can be switched off by SK operation in "Diagnosis control". It is also possible to wait for independent deactivation of the diagnosis program by the remote diagnosis central control unit.
2363 REMOTE DIAGNOSIS ACTIVE Programming of the EEPROMS can be performed only when remote diagnosis is not active.	* Switch off remote diagnosis.

ERROR CAUSE	ERROR REMEDY
2381 V.24 INIT-ERROR Error during initialization of the V24 board.	* Try again; HW reset or check board.
2677 WC BATTERY EXHAUSTED Battery of the word coupling card is exhausted.	* Replace battery (refer to information in "Interface conditions CC220/320").
2677 PC DUAL PORT BATTERY EMPTY PC dual port battery is empty.	* Replace battery (refer to information in "Interface conditions CC200/300").
2686 NO RAM WRITE PROTECTION No HW write protection is set on the MEM5 card. S1: 1=RAM; 2=EPROM.	* Activate write protection on the MEM5 card (refer to Page 7-1).
2998 ILL. PARAM IN UTILITY System warning (refer to Page 7-1). Internal data error, wrong transfer parameters.	* Inform after-sales service.
3009 KS ACCESS PLC blocks access to the KS table.	* Check PLC program.
3111 SAFETY CUT-OUT Safety cut-out during block search selection. With active safety cut-out; trying to start the block search.	* The block search starts automatically after termination of safety cut-out. * The block search can be selected again after termination of the safety cut-out.
3395 SPINDLE STOP The spindle has stopped, although a traversing movement is programmed (G95 is active or jogging in mm/rev. is selected).	* Make spindle run, e.g. program S-word.
3549 EDGE NOT RECOGNIZED In G875 the programmed end point was reached and the height-regulated axis has not covered the distance programmed with address S.	* Check NC program.
4639 DNC APPLIED SEVERAL TIMES Several DNC versions are applied simultaneously.	* Apply only one DNC version.
4640 DNC-INIT ERROR LSV2 interface cannot be initialized.	* HW reset (check whether the interface board is correct).

ERROR CAUSE	ERROR REMEDY
4641 DNC HARDWARE ERROR LSV2 interface cannot be initialized since the interface board signals a HW error.	* Check interface board.
4645 DNC INTERFACE MISSING LSV2: Wrong interface board.	* Replace interface board. * Inform after-sales service.
4648 DNC: MEMORY BLOCKED The LSV2 data file cannot be created.	* Wait until memory is freed up.
4649 DNC: MEMORY FULL The LSV2 data file cannot be created.	* Create memory space.
4652 ADDRESS 0 SET The user address is set to 0 on the BAB board.	* Reset user address.
4700 DNC AUTO.: NO BLOCKS No new data have been made available by the master control computer and the old data have already been processed.	* – reduce feed; – start with automatic execution only when the DA buffer is completely full; – increase the size of the DA buffer.
4701 NO PLC ACKNOWLEDGEMENT In the case of PLC word coupling, an internal check is made at the start of recording in the diagnostic programs D21 and D22 to establish whether the data can be transferred via the communication memory. If this is not possible, this message is issued and recording of PLC signals takes place via the interface.	* If transfer is to take place via the communication memory, a corresponding function module must be present in the PLC. Check correct function or presence of module.
4717 NO ACCESS TO DNC RAM Interface RAM is not released by the V24 board.	* Switch DNC off and on. * HW reset or check board.
4734 OUTPUT REFUSED/COMPUTER The output attempt was refused or aborted by the computer.	* Cause for refusal is in computer.
4735 READING IN REFUSED/COMPUTER The reading-in attempt was refused or aborted by the computer.	* Cause for refusal is in computer.

ERROR CAUSE	ERROR REMEDY
4754 STATUS TRANSFER System warning (refer to Page 7-1). Error during call of list management.	* Inform after-sales service.
4778 INVALID BAUD RATE Invalid baud rate for interface 2 on the SCP card.	* Set correct baud rate (see MPP P5500).
4804 KS ACCESS BLOCKED Display, search or store in KS table editor. Disturbance compensation: The PLC is to transfer values from the KS table to the servo-processor but does not release the KS table.	* Inform after-sales service. * Change PLC program.
4825 KS SYS: LIST MISSING System warning (refer to Page 7–1). It was not possible to delete the list.	* Control reset.
4826 KS SYS: LIST EXISTS System warning (refer to Page 7–1). Address-oriented reading-in of KS tables: The list for the search data is present before the start of reading-in.	* Control reset.
4827 KS SYS: PARAMETER System warning (refer to Page 7–1). Internal system error, call of a procedure with the wrong parameter.	* Control reset.
5203 BITBUS ASYNCHRONOUS Error while evaluating telegrams.	* Internal CNC problem: CLEAR LOGIC, possible inform after-sales service.
5290 BITBUS KS TABLE MISSING At least one KS table is missing that is required by the bitbus.	* Read in missing KS table.
5421 BITBUS BUFFER FULL Bitbus cannot process all telegrams.	* Make sure bitbus slaves are working correctly; if not, inform after-sales service.
5427 TELEGRAM UNKNOWN CNC is receiving “unknown” telegram through bitbus.	* Send only agreed telegrams.

ERROR CAUSE	ERROR REMEDY
5537 BITBUS ANSWER MISSING Slave is not responding to bitbus telegram.	* Identify missing telegram with bitbus diagnosis and eliminate cause.
5694 INVALID KS ADDRESS Trying to send a non-existent KS group as a telegram.	* Check KS table. * Check SD (105/106/107) for invalid values.
5695 TELEGRAM FAULTY Bitbus telegram is faulty.	* Identify faulty telegram with bitbus diagnosis and eliminate cause.
5696 BITBUS NEGATIVE ACKNOWLEDGE Slave is refusing telegram due to error.	* Check bitbus telegram.
5697 BITBUS RAM ACCESS No access to bitbus card is possible.	* Check bitbus card.
5698 SLAVE NOT RECEIVING Slave is not accepting any telegrams.	* Check slaves.
5699 SLAVE NOT AVAILABLE Master cannot send any telegrams to slave.	* Check slaves bitbus card and master-slave connecting cable; HW reset.
5700 BITBUS DRIVER INACTIVE The bitbus driver is not active.	* 1. HW reset 2. wait 10 seconds 3. check the bitbus card.
5704 BITBUS WAITING The bitbus is waiting too long for the axes to get into position.	* Check synchronous axes.

8. PANEL ERRORS

The control recognizes and displays errors in the panel and in the area of the panel interface.

ERROR CAUSE	ERROR REMEDY
2641 NO PANEL TRANSFER No connection between CP and panel. +24V logic voltage too low.	* Check connection between CP and panel. * Check resistance of connecting cable. * Check connector X12 at the CP and panel (pin inserted, wire broken?). * Interface on the CP faulty? * Interface module on panel control subassembly faulty. * Check + 24V on PS100 power supply unit (– 15%/+30%). * Is red LED lit on PS100 power supply unit?
2642 MTB I/O BOARD MISSING No connection between panel and MTB I/O subassembly. Interface module on the panel control subassembly faulty. Interface module on the MTB I/O faulty.	* Check connection between MTB I/O and panel. * Check resistance of the connecting cable. * Check connector X14 on the MTB I/O and panel (pin inserted, wire broken?). * Check +24V voltage supply at connector X10 on the MTB I/O subassembly. * Check fuse F1 (M0.5A) on the MTB I/O subassembly. * Replace panel control subassembly. * Replace MTB I/O subassembly.
2643 GRAPHIC BOARD MISSING No connection between graphic subassembly and panel control subassembly. +5V voltage for graphic subassembly missing. HW fault on the graphic subassembly. HW fault on the panel control subassembly.	* Check connection between graphic subassembly and panel control subassembly. * Check contacts at contact X16 on panel control subassembly. * Check +5 V at connector X3 (cold-soldered joint). * Replace power supply unit PSP. * Replace graphic subassembly. * Replace panel control subassembly.
2644 INVALID GRAPHIC PARAMETERS Graphic command wrong.	* Check transmission link between graphic subassembly and panel control. * Graphic subassembly faulty. Replace subassembly. * Interface module on panel control subassembly defective. Replace subassembly. * Interface module on CP subassembly defective. Replace subassembly.

ERROR CAUSE	ERROR REMEDY
2645 GRAPHICS HARDWARE ERROR Graphic command wrong.	* Check graphics HW. Replace subassembly.
2646 UART ERROR PANEL Interface of connector X12 at panel defective. Baud rate, handshake signal error. MPP, wrong parameters.	* Replace panel control subassembly. * Check baud rate, HS signals; change if necessary. * Check MPP; change if necessary.
2647 CLOCK POWER-FAIL Battery voltage too low. Clock not set after battery replacement. HW fault on CP subassembly.	* Check battery on the CP subassembly and voltage. Replace battery if necessary; set clock; error message is cleared. * Check contacts of battery holder (oxidation). * Replace CP subassembly.
2648 NC INTERFACE ERROR CP interface faulty (interface processor, interface RAM or EPROM).	* Interface software on CP subassembly defective or wrong. Replace EPROM.
2649 GRAPHIC INTERFACE ERROR Interface on panel control subassembly defective.	* Replace panel control subassembly. Refer to error 2643.
2650 MTB I/O-COUPPL. FAULTY MTB I/O coupling on panel control subassembly faulty.	* Replace panel control subassembly. Refer to error 2642.
2651 ERROR IN COMMAND Occurs in the event of control panel switch over to (ext. panel) CCS station.	* Check transmission link ext. panel-int. panel-CNC.
2652 PANEL NOT PRESENT The panel is not present with the specified No. in the system in the case of multiple panel operation.	* Check correct position of selector switch on each panel.
2653 NC NOT PRESENT The NC with the specified No. is not present in the system in the case of multiple NC operation.	* Check correct position of selector switch on each CP.

8.1 Error messages of the panel

Various hardware tests are performed in the panel when the panel is switched on and during operation. An error message is displayed on the screen if an error is detected.

ERROR CAUSE	ERROR REMEDY
> PANEL: EXT-MEMORY-ERROR AT ADDRESS = XXXX An error is established during the internal panel test. XXXX designates the errored RAM address. The panel cannot be operated.	* Inform after-sales service.
> NO DATA TRANSFER TO NC INTERFACE The message is output after a partyline command has been repeated 10 times. No data transfer to the NC Interface is possible.	* Control is switched off – switch on. * Control is faulty. * The partyline software is <u>not</u> compatible with the panel software. * Check NC transmitter line and connections.
> PANEL INACTIVE P-NO. = XX The panel is disconnected from the party-line (user number not 0 during power-up). The message is generated during operation with active graphic mode in order to inform the operator that it is a parallel panel. Only one display is possible.	* Activate the panel.
> PANEL TRANSMITTER LINE FAULTY! The PLC on the CP board cannot establish any contact with the panel, but receives data from the panel. The error is detected internally and the message output.	* Check transmitter line and connections.
> PANEL TRANSMITTER LINE FAULTY! An error is detected internally on the transmitter line to the PLC. This may be caused by a cable break or faulty modules in the PLC or panel.	* Check components.
> MTB I/O COUPLING HW-ERROR DATA= XXH Interface error to the MTB-I/O.	* Check interfaces on panel and MTB-I/O.
> GRAPHIC INTERFACE ERROR E-NO = XX Error in the graphics interface with indication of error code.	* Replace panel.

ERROR CAUSE	ERROR REMEDY
> NC INTERFACE: BI-PORT ERROR (HALT) The PLC establishes an error in the BI-port to P1. The PLC is now in an endless loop and can be activated again only by switching off the control. It is possible that this error will not occur again after this.	* Inform after-sales service if necessary and replace CP board.
> NC-INTERFACE: STATUS-ERROR (HALT) Error in the status port. The PLC is now in an endless loop and can be activated again only by switching off the control. It is possible that this error will not occur again after this.	* Inform after-sales service if necessary and replace CP board.
> NC-INTERFACE: CRT-RAM-ERROR (HALT) Error in the CRT-RAM. The PLC is now in an endless loop and can be activated again only by switching off the control. It is possible that this error will not occur again after this.	* Inform after-sales service if necessary and replace CP board.
> NC-INTERFACE: EXTERNAL-RAM-ERROR (HALT) Error in the data RAM. Data exchange can now no longer take place. The PLC is in an endless loop and can be activated again only by switching off the control. It is possible that this error will not occur again after this.	* Inform after-sales service if necessary and replace CP board.

9. CUSTOMER AREA ERRORS

In the CC220/320 controls, an EPROM area on the 3rd software module (CC320) or 1st software module (CC220) is designated as the “customer area”, which can be equipped by the user with his own software (MPP, texts, CPL programs, cycles etc.). The customer EPROM area is automatically recognized using an internal identifier at the start of each EPROM pair during control start-up and is thus integrated correspondingly in the system. The start-up cannot be completed in the event of a customer area error, since no parameters or texts exist in the customer area, for example. The corresponding customer EPROMs must therefore be supplemented before a new startup.

ERROR CAUSE		ERROR REMEDY
1 TEXT FILE MISSING		
No texts are present for the start-up.		* Install or add customer EPROM with valid texts in the customer area.
2 MACHINE PARAMETERS MISSING		
It is established during the start-up that no MPP is present in the customer EPROM area.		* Install or add customer EPROM with MPP in the customer area.
3 CUSTOM. EPROM ON CODE AREA		
During the start-up, it is established that customer EPROMs are present in the code area of the 3rd software module.		* Install original EPROM again.

